# A BKQR Level-0 Critical Representation Criterion for the Three-Dimensional Navier–Stokes Equations

14 August 2026

*Internal research preprint — conditional calibration theorem*

**Abstract**

We formulate a conservative "Level-0" criterion for the three-dimensional incompressible Navier–Stokes equations on the periodic torus in terms of a geometric family of positive BKQR regularizations. Let $u_h$ denote the solution of a fixed-$h$ regularized equation driven by the positive Stokes multiplier $J_h = f_q(hA)$, where $0 < q < 1$ and $f_q(qz) = (1+z)f_q(z)$. Along the geometric ladder $h_n = h_* q^n$, assume square summability, at the critical weight $h_n^{-1/2}$, of both neighboring representation increments $u_{h_{n+1}} - u_{h_n}$ and clock defects $(I - J_{h_n})u_{h_n}$ in $C_t L^2_x$. We show that the representation orbit converges strongly to a solution $u$ of the unregularized Navier–Stokes equations and that a discrete BKQR clock square function reconstructs the homogeneous $H^{1/2}$ norm of $u$. Consequently $u \in L^\infty_t H^{1/2}_x \hookrightarrow L^\infty_t L^3_x$, so the standard endpoint regularity criterion applies. The result is deliberately not stronger than the classical critical endpoint; its purpose is to provide a rigorous calibration point for weaker BKQR Levels 1–4. In particular, the theorem does not prove that the Level-0 hypothesis holds for arbitrary three-dimensional data.

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# 1 Introduction

The three-dimensional incompressible Navier–Stokes system on the periodic torus is

$$\partial_t u + \nu A u + B(u,u) = 0, \qquad \nabla \cdot u = 0, \tag{1.1}$$

where $A = -P\Delta$ is the positive Stokes operator and

$$B(a,b) = P\nabla \cdot (a \otimes b) = P((a \cdot \nabla)b). \tag{1.2}$$

Here $P$ is the Leray projector and $\nu > 0$ is the viscosity.

The BKQR framework introduces, independently of physical time, a positive representation scale $h > 0$ and a fixed geometric ratio $q \in (0,1)$. The principal purpose of Level 0 is not to enlarge the known regularity class, but to answer a more basic calibration question: *what representation-space control is sufficient to reconstruct the standard critical $H^{1/2}$ endpoint directly from the regularized BKQR orbit?*

The answer is a critical square-variation condition on the geometric orbit $h \mapsto u_h$. The argument has four structural ingredients:

(i) positivity and energy control for the fixed-$h$ regularized family;

(ii) a geometric Hardy inequality converting neighboring increments into control of the full representation tail;

(iii) a discrete clock square-function identity equivalent to $\dot{H}^{1/2}$;

(iv) strong compactness sufficient to pass the regularized convection to the physical Navier–Stokes nonlinearity.

The criterion is conditional. It should therefore be read as a theorem of the form

$$\text{critical representation control} \quad \Longrightarrow \quad \text{standard critical physical regularity},$$

not as a proof that the critical representation control is automatic for arbitrary data.

# 2 Functional setting and the positive BKQR clock

Let

$$H = L^2_\sigma(\mathbb{T}^3)$$

be the mean-zero divergence-free subspace of $L^2(\mathbb{T}^3;\mathbb{R}^3)$. The Stokes operator $A = -P\Delta$ is positive and self-adjoint on $H$, with compact inverse and first positive eigenvalue $\lambda_1 > 0$. We use the standard Hilbert scale generated by $A$; in particular $D(A^{1/4}) = H^{1/2}_\sigma$ up to equivalent periodic norms.

Fix $q \in (0,1)$ and define

$$f_q(z) = \prod_{m=0}^{\infty} (1 + q^m z)^{-1}, \qquad z \geq 0. \tag{2.1}$$

The elementary self-similarity identity

$$f_q(qz) = (1+z) f_q(z) \tag{2.2}$$

is exact. By spectral functional calculus define

$$J_h = f_q(hA). \tag{2.3}$$

Then

$$J_{qh} = (I + hA) J_h. \tag{2.4}$$

Since $0 < f_q(z) \leq 1$ for $z \geq 0$, every $J_h$ is a positive self-adjoint contraction on every Sobolev space in the Stokes scale.

*Hypothesis* 2.1 (Fixed-$h$ regularized family). For each $h > 0$ there exists a sufficiently regular divergence-free solution $u_h$ on $[0,T]$ of

$$\partial_t u_h + \nu A u_h + B(J_h u_h, u_h) = 0, \qquad u_h(0) = u_0, \tag{2.5}$$

with the exact energy identity

$$\frac{1}{2} \|u_h(t)\|_2^2 + \nu \int_0^t \left\| A^{1/2} u_h(s) \right\|_2^2 \, \mathrm{d}s = \frac{1}{2} \|u_0\|_2^2 . \tag{2.6}$$

The energy identity follows formally from divergence-freeness of $J_h u_h$ and is one of the frozen structural inputs of the BKQR regularization. Publication-level treatment of the exact fixed-$h$ well-posedness theorem is logically separate from the Level-0 argument.

## 3 The Level-0 representation criterion

Choose a coarse scale $h_* > 0$ and the geometric ladder

$$h_n = h_* q^n, \qquad n \geq 0. \tag{3.1}$$

We assume for simplicity that

$$h_* \lambda_1 \geq 1. \tag{3.2}$$

This ensures that every Stokes frequency crosses a clock transition scale within the ladder. The assumption can be removed by adding the harmless $L^2$ low-frequency contribution explicitly.

Define the neighboring increments and clock defects

$$\delta_n = u_{h_{n+1}} - u_{h_n}, \qquad m_n = (I - J_{h_n}) u_{h_n}. \tag{3.3}$$

**Definition 3.1** (Level-0 representation seminorm)**.** Set

$$\mathfrak{R}_0(T)^2 := \sum_{n=0}^{\infty} h_n^{-1/2} \left[ \|\delta_n\|_{C([0,T];L^2)}^2 + \|m_n\|_{C([0,T];L^2)}^2 \right]. \tag{3.4}$$

The defining Level-0 assumption is

$$\mathfrak{R}_0(T) < \infty. \tag{3.5}$$

It directly pays for the critical representation motion. No shear, helicity, null-cone, quotient, or higher covariance cancellation is used.

# 4 Convergence of the geometric representation orbit

**Lemma 4.1** (Absolute summability of neighboring increments)**.** *Under* (3.5)*,*

$$\sum_{n=0}^{\infty} \|\delta_n\|_{C_t L^2_x} < \infty. \tag{4.1}$$

*Consequently $u_{h_n}$ converges strongly in $C([0,T]; L^2_\sigma)$ to a limit $u$.*

*Proof.* By Cauchy–Schwarz,

$$\sum_n \|\delta_n\|_{C_t L^2_x} \le \left(\sum_n h_n^{1/2}\right)^{1/2} \left(\sum_n h_n^{-1/2} \|\delta_n\|^2_{C_t L^2_x}\right)^{1/2}. \tag{4.2}$$

The first factor is finite because $h_n = h_* q^n$. Hence the telescoping series converges uniformly in time with values in $L^2$. □

The next lemma retains the critical weight after summing the entire tail.

**Lemma 4.2** (Geometric Hardy estimate)**.** *Let $a_n \ge 0$. Then*

$$\sum_{n=0}^{\infty} h_n^{-1/2} \left(\sum_{m=n}^{\infty} a_m\right)^2 \le \frac{1}{(1-q^{1/4})^2} \sum_{n=0}^{\infty} h_n^{-1/2} a_n^2. \tag{4.3}$$

*In particular,*

$$\sum_n h_n^{-1/2} \|u - u_{h_n}\|^2_{C_t L^2_x} \le C_q \sum_n h_n^{-1/2} \|\delta_n\|^2_{C_t L^2_x}. \tag{4.4}$$

*Proof.* Since $h_{n+r} = h_n q^r$,

$$h_n^{-1/4} \sum_{m \ge n} a_m = \sum_{r \ge 0} q^{r/4} h_{n+r}^{-1/4} a_{n+r}.$$

Apply Young's convolution inequality on $\ell^2(\mathbb{N}_0)$ with kernel $q^{r/4}$, whose $\ell^1$ norm is $(1 - q^{1/4})^{-1}$. □

# 5 The critical BKQR clock square function

The decisive analytic statement is that the discrete collection of clock defects reconstructs exactly one half spatial derivative.

**Proposition 5.1** (Clock square-function equivalence)**.** *Assume* (3.2)*. There exist constants $0 < c_q \le C_q < \infty$, depending only on $q$ and the normalization of $h_*$ relative to $\lambda_1$, such that*

*for every mean-zero $v \in D(A^{1/4})$,*

$$c_q \left\|A^{1/4}v\right\|_2^2 \le \sum_{n=0}^{\infty} h_n^{-1/2} \left\|(I - J_{h_n})v\right\|_2^2 \le C_q \left\|A^{1/4}v\right\|_2^2. \tag{5.1}$$

*Conversely, finiteness of the middle expression implies $v \in D(A^{1/4})$.*

*Proof.* By the spectral theorem it suffices to prove the scalar estimate

$$S(\lambda) := \sum_{n=0}^{\infty} h_n^{-1/2} |1 - f_q(h_n\lambda)|^2 \asymp_q \lambda^{1/2}, \qquad \lambda \ge \lambda_1. \tag{5.2}$$

Choose $n_*$ so that $1 \le h_{n_*}\lambda < q^{-1}$. On the compact interval $[1, q^{-1}]$, $1 - f_q$ is bounded below by a positive $q$-dependent constant; hence

$$S(\lambda) \ge c_q h_{n_*}^{-1/2} \asymp_q \lambda^{1/2}.$$

For $n \le n_*$, use $|1 - f_q(x)| \le 1$. Since $h_n^{-1/2}$ is geometric, the corresponding partial sum is bounded by $C_q h_{n_*}^{-1/2} \lesssim_q \lambda^{1/2}$.

For $n > n_*$ we have $h_n\lambda < 1$. From the product representation,

$$0 \le 1 - f_q(x) \le C_q x, \qquad 0 \le x \le 1. \tag{5.3}$$

Therefore

$$\sum_{n>n_*} h_n^{-1/2} |1 - f_q(h_n\lambda)|^2 \lesssim_q \lambda^2 \sum_{n>n_*} h_n^{3/2} \lesssim_q \lambda^{1/2}.$$

This proves (5.2); integration against the spectral measure of $A$ yields (5.1). □

We now transfer the clock defects from the regularized family to the physical limit.

**Lemma 5.2** (Clock defects of the limit)**.** *Under* (3.5)*,*

$$\sum_n h_n^{-1/2} \left\|(I - J_{h_n})u\right\|_{C_t L_x^2}^2 \le C_q \mathfrak{R}_0(T)^2. \tag{5.4}$$

*Proof.* Since $0 \le J_h \le I$ spectrally,

$$\left\|(I - J_{h_n})u\right\|_2 \le \left\|u - u_{h_n}\right\|_2 + \left\|(I - J_{h_n})u_{h_n}\right\|_2.$$

Square, sum with weight $h_n^{-1/2}$, and use Lemma 4.2 and the definition of $\mathfrak{R}_0(T)$. □

**Corollary 5.3** (Critical physical control)**.** *Under the Level-0 hypothesis,*

$$\sup_{0\le t\le T} \left\|A^{1/4}u(t)\right\|_2^2 \le C_q \mathfrak{R}_0(T)^2. \tag{5.5}$$

*Hence*

$$u \in L^\infty(0,T; H^{1/2}_\sigma(\mathbb{T}^3)) \hookrightarrow L^\infty(0,T; L^3(\mathbb{T}^3)). \tag{5.6}$$

# 6 Passage to the physical Navier–Stokes equation

The Level-0 criterion must also identify the limit $u$ as a solution of the unregularized equation, rather than merely as a critical function.

**Lemma 6.1** (Strong $L^2_t L^3_x$ convergence)**.** *The sequence $u_{h_n}$ converges strongly to $u$ in $L^2(0,T;L^3(\mathbb{T}^3))$.*

*Proof.* The energy identity gives a uniform bound in $L^2(0,T;H^1)$. By Lemma 4.1, $u_{h_n} - u \to 0$ in $L^\infty(0,T;L^2)$. Interpolation between $L^2$ and $L^6$ gives

$$\|v\|_{L^3}^2 \lesssim \|v\|_{L^2} \|v\|_{L^6}.$$

Hence

$$\|u_{h_n} - u\|_{L^2_t L^3_x}^2 \lesssim T^{1/2} \|u_{h_n} - u\|_{L^\infty_t L^2_x} \|u_{h_n} - u\|_{L^2_t L^6_x},$$

and the last factor is uniformly bounded. □

**Lemma 6.2** (Convergence of the advecting velocity)**.** *One has*

$$J_{h_n} u_{h_n} \longrightarrow u \quad \textit{strongly in } L^2(0,T;L^3(\mathbb{T}^3)). \tag{6.1}$$

*Proof.* Decompose

$$J_{h_n} u_{h_n} - u = J_{h_n}(u_{h_n} - u) + (J_{h_n} - I)u.$$

The first term converges to zero in $L^2_t L^3_x$ by Lemma 6.1 and the contractivity of $J_h$ on $L^2$ and $H^1$, followed by interpolation. The limit $u$ belongs to $L^2(0,T;H^1)$ by weak lower semicontinuity of the energy bound. Since $f_q(h\lambda) \to 1$ for each spectral value and $|f_q(h\lambda)| \le 1$, dominated convergence in the spectral representation yields

$$(J_h - I)u \to 0 \quad \text{strongly in } L^2(0,T;H^1),$$

which proves the claim. □

**Proposition 6.3** (Limit equation)**.** *The limit $u$ is a distributional, energy-class solution of (1.1) with $u(0) = u_0$.*

*Proof.* By Lemmas 6.1–6.2,

$$(J_{h_n} u_{h_n}) \otimes u_{h_n} \to u \otimes u \quad \text{strongly in } L^1(0,T;L^{3/2}).$$

The viscous term passes to the limit using the weak $L^2_t H^1_x$ convergence. Thus the weak formulation of (2.5) converges to the weak formulation of (1.1). Since the convergence in Lemma 4.1 is uniform in time with values in $L^2$, the initial trace is preserved: $u(0) = u_0$. □

# 7 Main theorem

**Theorem 7.1** (BKQR Level-0 critical representation criterion)**.** *Let $u_0$ be smooth, mean-zero and divergence-free on $\mathbb{T}^3$, and suppose Hypothesis 2.1 holds for the positive BKQR regularized family. Let $h_n = h_* q^n$ with $h_* \lambda_1 \ge 1$. If*

$$\sum_{n=0}^{\infty} h_n^{-1/2} \left[ \|u_{h_{n+1}} - u_{h_n}\|_{C([0,T];L^2)}^2 + \|(I - J_{h_n})u_{h_n}\|_{C([0,T];L^2)}^2 \right] < \infty, \tag{7.1}$$

*then the sequence $u_{h_n}$ converges strongly in $C([0,T];L^2)$ to an exact solution $u$ of the unregularized three-dimensional Navier–Stokes equations, and*

$$u \in L^\infty(0,T;H^{1/2}_\sigma(\mathbb{T}^3)) \hookrightarrow L^\infty(0,T;L^3(\mathbb{T}^3)). \tag{7.2}$$

*Consequently the standard endpoint $L^\infty_t L^3_x$ regularity theorem upgrades $u$ to the strong regular solution on $[0,T]$. If (7.1) holds on every finite interval, regularity extends through every finite time.*

*Proof.* Strong convergence in $C_t L^2_x$ follows from Lemma 4.1. Lemmas 4.2 and 5.2, combined with Proposition 5.1, give (7.2). Proposition 6.3 identifies the limit as a solution of the physical Navier–Stokes equation with the correct initial data. The final implication is the standard critical endpoint regularity criterion. □

# 8 Compatibility with the Candidate-F residual

Although Theorem 7.1 already reaches the critical endpoint directly, it is useful to verify that the same hypothesis also lands naturally in the exact BKQR Candidate-F residual architecture.

Let

$$U_K = \sum_{j=0}^{K} c_j^{(K)} u_{h_j}, \qquad d_j = u_{h_j} - U_K,$$

where the BKQR coefficients satisfy $\sum_j c_j^{(K)} = 1$ and are uniformly bounded in $\ell^1$ for fixed $q$. The exact Candidate-F identity is

$$\partial_t U_K + \nu A U_K + B(U_K, U_K) = P\nabla \cdot G_K, \tag{8.1}$$

with

$$G_K = -\sum_{j=0}^{K} c_j^{(K)} d_j \otimes d_j + \sum_{j=0}^{K} c_j^{(K)} [(I - J_{h_j})u_{h_j}] \otimes u_{h_j}. \tag{8.2}$$

All first-order representation disagreement cancels exactly.

**Corollary 8.1** (Candidate-F landing under Level 0)**.** *Under the hypotheses of Theorem 7.1, the neighboring increments satisfy*

$$\delta_n \to 0 \quad \textit{strongly in } L^2(0,T;L^3),$$

*and the clock defects satisfy the same conclusion. Consequently the dispersion and mollifier parts of (8.2) vanish in $L^1(0,T;L^{3/2})$ along the deep geometric ladder, in the sense of the frozen no-rate Candidate-F tail-continuity theorem.*

*Proof.* The $C_t L^2$ convergence of $\delta_n$ and uniform $L^2_t H^1_x$ bounds imply $\delta_n \to 0$ in $L^2_t L^3_x$ exactly as in Lemma 6.1. For $m_n = (I - J_{h_n})u_{h_n}$, the Level-0 sum gives $m_n \to 0$ in $C_t L^2_x$, while spectral contractivity gives a uniform $L^2_t H^1_x$ bound; hence $m_n \to 0$ in $L^2_t L^3_x$. Products therefore converge to zero in $L^1_t L^{3/2}_x$. The no-rate Candidate-F theorem then handles the signed BKQR aggregation. □

## 9 Interpretation and limitations

Theorem 7.1 is intentionally the strongest-assumption, safest member of the proposed BKQR hierarchy. Its main significance is structural rather than an enlargement of the classical solution class.

**Level 0 is a calibration theorem.** The conclusion $L_t^\infty H_x^{1/2} \hookrightarrow L_t^\infty L_x^3$ lies in standard critical territory. The criterion therefore does not, by itself, extend the known endpoint regularity region.

**The hypothesis is not presently known to be automatic.** The unresolved three-dimensional difficulty has merely been transferred into the question of whether the critical representation sum (7.1) can be established from weaker, non-circular information. Higher BKQR levels are designed precisely to relax that demand by exploiting exact null fibres, near-null geometry, individual quadratic depletion, or assembled covariance cancellation.

**No universal $h^\alpha$ law is assumed.** The criterion is a weighted square-summability statement. It does not require a pointwise power estimate of the form $\|u_{qh} - u_h\| \lesssim h^\alpha$, a route known to be too strong at the moving face.

**Status separation.** The exact BKQR clock identity, positive functional calculus, Candidate-F quadraticization, and coefficient stability belong to the frozen algebraic core. The fixed-$h$ global theory for the precise regularization and the standard endpoint theorem are neighboring PDE inputs. The new content of Level 0 is the assembly of these ingredients with the clock square-function and geometric Hardy estimate into a single explicit conditional criterion.

## 10 Proof-audit checklist for a publication version

Before treating the present note as publication-ready, the following items should be frozen in full detail:

1. a precise fixed-$h$ well-posedness theorem for (2.5) in the spaces used above;
2. a complete proof of Proposition 5.1, with constants and the low-frequency normalization stated uniformly;
3. the limit passage in Proposition 6.3 in one fixed weak-solution framework;
4. a precise statement and citation of the endpoint $L_t^\infty L_x^3$ regularity theorem in the periodic setting;
5. a publication-level statement of the no-rate Candidate-F tail-continuity theorem used in Corollary 8.1.

None of these items should be confused with a proof that (7.1) holds for arbitrary data.

## 11 Position in the BKQR hierarchy

The role of the successive levels may be summarized as follows:

| Level | admissible representation motion |
|---|---|
| 0 | full critical representation motion must be summably controlled; |
| 1 | large motion is allowed inside exact nonlinear-null fibres; |
| 2 | large motion may decompose into weakly interacting null packets; |
| 3 | only each individual quadratic physical defect must be small; |
| 4 | only the assembled physical covariance/residual must be small. |

Thus Level 0 serves as the baseline against which the genuine enlargement produced by Levels 1–4 must be measured.

## References and internal provenance

[1] *BKQR / 3D Navier–Stokes — Critical Context for Reuse: Left Path + Current Audited Core*, internal research ledger, 14 August 2026.

[2] *BKQR / 3D Navier–Stokes — Critical Context Transfer for a New Prompt*, internal research summary, 14 August 2026.

[3] *BKQR / 3D Navier–Stokes — Critical Context Handoff*, internal research ledger, 14 August 2026.

[4] Standard endpoint regularity theorem for suitable/energy-class three-dimensional Navier–Stokes solutions in $L_t^\infty L_x^3$; exact bibliographic citation to be frozen in a publication version.

## Source-context appendices

The following pages reproduce the project source/context material that accompanied the Level-0 derivation. They are retained as an internal provenance and audit appendix. Their inclusion does not promote statements labelled *open*, *conditional*, *derived/audit*, or *falsified* to theorem status. The formal paper above should be read according to the status distinctions stated explicitly in the source ledger.

*Verbatim source-context pages follow.*

# Appendix A - Current Critical Context

Raw project-context snapshot appended for provenance and reuse. It may contain duplicated material, research-status labels, and open/falsified branches; it is not promoted by inclusion into the Level-0 theorem.

### BKQR / 3D NAVIER-STOKES — CRITICAL CONTEXT FOR REUSE

### LEFT PATH + CURRENT AUDITED CORE

Date: 2026-08-14

### PURPOSE

This file is a compact prompt-context summary for continuing the BKQR/Navier–Stokes research in a new chat. It is NOT a proof of 3D Navier–Stokes regularity. Keep exact/frozen results separate from derived/audit results and open theorems.

### 0. MINIMAL SETTING

Domain:

    Ω = T^3
    div u = 0
    ν > 0
    A = -PΔ  (positive Stokes operator)
    B(u,v) = P div(u ⊗ v)

Physical NSE:

    ∂_t u + ν A u + B(u,u) = 0

Representation scale:

    h > 0
    fixed geometric ratio 0 < q < 1 (often q = 1/2)
    (S F)(h) = F(qh)

Physical time is t. h and K are representation-scale variables.

### 1. FROZEN BKQR SCALE ALGEBRA

Geometric annihilator:

    P_K(S) = Π_{m=1}^K (S - q^m)/(1 - q^m)
           = Σ_{j=0}^K c_j^(K) S^j

Exact identities:

    Σ_j c_j = 1
    Σ_j c_j q^(jn) = 0, 1 <= n <= K
    sup_K Σ_j |c_j| < ∞  for fixed q

Correction layers:

    C_0 = I
    C_K = P_K - P_{K-1}

Exact q-product / bilinear lift:

    C_K Q(F,G)(h)
      = Σ_{r=0}^K Q(C_r F(q^(K-r)h), C_{K-r} G(h))

for any h-independent bilinear map Q.

This makes the correction tower a multiplication-aware q-convolution algebra.

Boundary-vanishing ideals multiply:

    Q(I_μ, I_ν) ⊂ I_{μν}

In particular:

    Q(I_μ, I_μ) ⊂ I_{μ^2}

### 2. POSITIVE BKQR CLOCK AND SHELLS

Clock kernel:

    f_q(z) = Π_{n>=0} (1 + q^n z)^(-1)

Self-similar identity:

    f_q(qz) = (1+z) f_q(z)

Positive Stokes regularizer:

    J_h = f_q(hA)

Exact operator identity:

    J_{qh} = (I + hA) J_h

Probabilistic representation:

    J_h = E[ exp(-h Y_∞ A) ]

with Y_∞ = Σ_{n>=0} q^n E_n, E_n ~ Exp(1).

Positive q-shells:

    Π_K(hA) = C_K J_h >= 0
    Σ_K Π_K(hA) = I

Exact scalar shell multiplier:

$$\pi_K(x) = q^{K(K-1)/2} x^K f_q(x) / (q;q)_K$$

Exact raising relation:

$$hA\, \Pi_K(hA) = (q^{-K} - q)\, \Pi_{K+1}(hA)$$

Bernoulli shell law:

$$\Sigma_K t^K \pi_K(x) = f_q(x)/f_q(tx)$$

so K is a logarithmic spectral coordinate.

Mean shell-number function:

$$g(x) = E[N_x] = \Sigma_{n>=0} q^n x/(1+q^n x)$$

Shell-number operator:

$$N_h = g(hA)$$

Frozen exact/proved:

$$g(x) = -x\, \partial_x \log f_q(x)$$

$$x\, g'(x) = \mathrm{Var}(N_x)$$

$$0 <= x\, g'(x) <= C_q$$

$$|g(x)-g(y)| <= C_q\, |\log(x/y)|$$

Critical shell moment:

$$m_\sigma(x) = f_q(x)/f_q(B^\sigma x),\ B=q^{-1}$$

For $0 <= \sigma <= 1$:

$$m_\sigma(x) \asymp (1+x)^\sigma$$

At $\sigma = 1/2$:

$$M_{1/2}(u) = \langle u, m_{1/2}(hA) u \rangle$$

is equivalent to an inhomogeneous $H^{1/2}$ energy, hence controls $L^3$ on $T^3$.

## 3. POSITIVE REGULARIZED NSE FAMILY

Preferred regularized equation:

$$\partial_t u_h + \nu A u_h + B(J_h u_h, u_h) = 0$$

$$u_h(0) = u_0$$

For fixed $h > 0$ this is expected/standard-neighbor globally controlled.

Energy identity:

$$1/2\, d/dt\, ||u_h||_2^2 + \nu\, ||A^{1/2} u_h||_2^2 = 0$$

Reason:

$J_h u_h$ remains divergence-free.

## 4. EXACT CANDIDATE-F RESIDUAL

Sample:

$$h_j = h q^j$$

$$u_j = u_{h_j}$$

Extrapolated field:

$$U_K = \Sigma_{j=0}^K c_j u_j$$

Disagreements:

$$d_j = u_j - U_K$$

Exact affine cancellation:

$$\Sigma_j c_j d_j = 0$$

Exact residual:

$$\partial_t U_K + \nu A U_K + B(U_K, U_K) = P \,\mathrm{div}\, G_K$$

with

$$G_K = - \Sigma_j c_j d_j \otimes d_j + \Sigma_j c_j [(I-J_j)u_j] \otimes u_j$$

Write:

$$G_K = G_K^{disp} + G_K^{moll}$$

### CRITICAL FACT:

All first-order cross-scale disagreement cancels exactly. The inter-scale dispersion residual starts quadratically.

## 5. CANDIDATE-F GREEN MATRIX

Neighboring increments:

$$\delta_r = u_{r+1} - u_r$$

Then:

$$G_K^{disp} = - \Sigma_{r,s<K} M_{rs}^{(K)}\, \delta_r \otimes \delta_s$$

Exact Green kernel:

$$M_{rs}^{(K)} = p_{\min(r,s)}\, \tau_{\max(r,s)}$$

where

$$p_r = \Sigma_{j<=r} c_j$$

$$\tau_r = \Sigma_{j>r} c_j = 1-p_r$$

Exact inverse:

```
    (M^{(K)})^{-1}
      = D_K^* diag(c_0^{-1},...,c_K^{-1}) D_K
```

Important falsification:

```
    ||M^{(K)}||_op -> 0  is FALSE.
```

M^{(K)} is full rank and indefinite.

Correct replacement: The non-small part travels to the finest-scale corner.

Universal moving corner: After reversing indices, M^{(K)} converges in trace/operator norm to a nonzero universal kernel M_*.

Weighted finest-corner contraction: For D_ρ = diag(1,ρ,...,ρ^{K-1}), 0<ρ<1,

```
    ||D_ρ M^{(K)} D_ρ|| <= C(q,ρ) ρ^{2K}
```

Strong no-rate consequence: If

```
    sup_r ||δ_r||_Y < ∞
```

and

```
    ||δ_r||_Y -> 0
```

then

```
    ||G_K^disp||_X -> 0
```

for every compatible bounded bilinear map Y×Y -> X.

Thus Candidate F does NOT require a universal power law

```
    ||u_{qh}-u_h|| ~ h^α.
```

Plain tail continuity is enough.

## 6. FROZEN SHEAR-NULL GEOMETRY

1D2C shear algebra:

```
    N_n = { W(x)=F(n·x) : F·n=0 }
```

2D1C shear algebra:

```
    N_v = { v φ(x) : v·∇φ=0 }
```

Exact:

```
    B(W_1,W_2)=0
```

for all W_1,W_2 in the same fixed shear algebra.

A and all spectral multipliers of A preserve each fixed shear algebra.

Therefore a common shear ladder gives zero physical Candidate-F forcing even though M_* is nonzero.

Local degenerate-C4 theorem: Persistent degenerate C4 geometry gives either

```
    (i) plane synchronization
```

or

```
    (ii) exact common-sector shear nullity.
```

Do NOT globalize this using the old “Global B2” statement; that globalization was falsified.

## 7. LEFT PATH: COVARIANT SCALE TANGENT

Raw logarithmic scale tangent:

```
    Z_h = -h ∂_h u_h
```

Clock derivative:

```
    K_h = -h ∂_h J_h
```

Frozen exact factorization:

```
    K_h = J_h N_h
```

Linearized regularized operator:

```
    L_h V
      = ∂_t V + ν A V
        + B(J_h V,u_h)
        + B(J_h u_h,V)
```

Raw tangent equation:

```
    L_h Z_h = -B(K_h u_h,u_h)
```

Define covariant tangent:

```
    C_h = Z_h + N_h u_h
```

### FROZEN EXACT COVARIANT CANCELLATION:

```
    L_h C_h = [ C_{J_h u_h}, N_h ] u_h
```

where C_a v := B(a,v).

This is the main LEFT-path equation.

It replaces direct clock forcing by a shell-number commutator.

## 8. FROZEN SHELL-NUMBER COMMUTATOR SYMBOL

For divergence-free a,

```
    [C_a,N_h]v
```

has Fourier triad symbol for k=p+r:

```
    i P_k
      (r · a_hat(p))
```

```
      [ g(h|r|^2) - g(h|k|^2) ]
      v_hat(r)
Exact equal-radius nullity:
    |r|=|k|  =>  contribution = 0
Frozen low-high derivative transfer: If |p| <= |r|/2, then
    |r| |g(h|r|^2)-g(h|k|^2)| <= C_q |p|
Thus the derivative moves from the high transported mode to the low advector.
```

## 9. NEW LEFT DERIVATION: THREE-FREQUENCY COVARIANT GAIN

```
For the actual advector a = J_h u_h, one also gets the clock factor f_q(h|p|^2).
Incompressibility gives for high-high -> low:
    p·u_hat(p)=0
    r=k-p
so
    r·u_hat(p) = k·u_hat(p)
Thus the derivative becomes the low OUTPUT frequency.
Derived/audit global symbol heuristic:
    |m_h(k,p,r)| <= C_q min{|k|,|p|,|r|}
after including the special J_h factor.
This is stronger than the generic shell-number commutator estimate.
```

## 10. PARABOLIC X_T / Y_T CANDIDATE SPACES

```
Dyadic frequencies:
    λ_j = 2^j
Parabolic time cells:
    I_{j,n}
      = [ n/(νλ_j^2), (n+1)/(νλ_j^2) ]
Candidate stress norm:
    ||G||_{X_T}^2
      = Σ_{j,n}
        [ λ_j^2
          ||Δ_j G||_{L^1(I_{j,n};L^{3/2})}
        ]^2
Candidate solution shell variables:
    A_j = λ_j^{1/2} ||Δ_j u||_{L_t^∞ L_x^2}
    B_j = ν^{1/2} λ_j^{3/2} ||Δ_j u||_{L_t^2 L_x^2}
Y_T is the corresponding critical combination of □^2(A_j) and □^2(B_j).
Full publication-level Galerkin-uniform mapping theorem is still to be audited.
```

## 11. DERIVED FAR-INTERACTION KERNELS

```
Split triads by dyadic gap m.
LOW ADVECTOR / HIGH TRANSPORTED -> HIGH:
    K_LH(m) <= C 2^{-m}
HIGH ADVECTOR / LOW TRANSPORTED -> HIGH:
    K_HL(m) <= C (1+m) 2^{-m/2}
HIGH-HIGH -> LOW:
    K_HHL(m) <= C (1+m) 2^{-2m}
The parabolic time cells are essential in HL and HHL.
Therefore:
    η_L
      := max tail □^2 norms of these kernels
      <= C_q (1+L) 2^{-L/2}
      -> 0
Derived/audit theorem candidate:
    ||G_h^{far,L}(a,u)||_{X_T}
      <= (C_q/ν) η_L ||a||_{Y_T} ||u||_{Y_T}
For a=J_h u_h:
    ||G_h^{cov,far,L}||_{X_T}
      <= (C_q/ν) η_L ||u_h||_{Y_T}^2
This should be treated as DERIVED / AUDIT until the full LP support/cell overlap proof is independently checked.
```

## 12. MOVING-FACE ENVELOPE

```
For comparable-frequency triads
    |p| ~ |r| ~ |k| ~ Λ
set
    x = h Λ^2
The full covariant coefficient contains:
```

```
    f_q(h|p|^2)
    [g(h|r|^2)-g(h|k|^2)]
Derived scalar envelope:
    coefficient <= C_{q,L} min{x,x^{-1}}
Reason: - below face x<<1: g(x)=O(x) - above face x>>1: f_q(x)<=1/(1+x)
Define finite-width face:
    2^{-M} <= hΛ^2 <= 2^M
Comparable triads outside the face gain:
    O(2^{-M})
Hence derived/audit:
    ||G_h^{local,offface;L,M}||_{X_T}
      <= (C_{q,L}/ν) 2^{-M} ||u_h||_{Y_T}^2
```

## 13. COVARIANT MOVING-FACE REDUCTION

```
Decompose:
    G_h^cov
      = G_h^{far,L}
        + G_h^{local,offface;L,M}
        + G_h^{face;L,M}
Then derived/conditional:
    ||G_h^cov - G_h^{face;L,M}||_{X_T}
      <= (C_q/ν) η_L ||u_h||_{Y_T}^2
        + (C_{q,L}/ν) 2^{-M} ||u_h||_{Y_T}^2
Choose L large, then M large.
Conclusion: Any nonperturbative LEFT-path response obstruction must live in
    |p| ~ |r| ~ |k|
and
    h|p|^2 ~ h|r|^2 ~ h|k|^2 ~ 1.
So the unresolved Q2 response is forced to a finite-width, comparable-frequency moving face.
This is currently the central LEFT reduction.
```

## 14. COVARIANT -> ORDINARY SCALE CONTINUITY

```
Define:
    R_h = (I+hA)^{-1}
    ε_h = u_{qh} - R_h u_h
Frozen exact parallel transport:
    ε_h
      = (log B)
        ∫_0^1
        m_{-(1-s)}(q^s h A)
        C_{q^s h} ds
Hence in a compatible topology:
    C_h -> 0
      => ε_h -> 0
      => u_{qh} - u_h -> 0
Then no-rate Candidate F gives:
    u_{qh}-u_h -> 0
      => G_K^disp -> 0
This is the downstream landing of the LEFT path.
```

## 15. CLOCK-FACTOR TANGENTS AND RADIAL SEPARATION

```
Factor:
    J_h = Π_{n>=0} (I+q^n hA)^(-1)
n-th factor tangent:
    K_{n,h}
      = J_h [ q^n hA / (I+q^n hA) ]
Scalar tangent:
    p_n(x)=q^n x/(1+q^n x)
Frozen finite-Galerkin radial separation: If a linear functional is invisible to all K_{n,h}, then cancellation
between distinct Stokes radii is impossible.
New finite Cauchy–Vandermonde sharpening: For M distinct active radii x_j=hλ_j, the M×M matrix
    P_{n j}=q^n x_j/(1+q^n x_j), 0<=n<M
has determinant proportional to
    Π_{i<j}(x_i-x_j)
Therefore M distinct radii are separated by M clock-factor directions.
The only radial degeneracy is radius collision.
```

## 16. TWO-CLOCK TRIAD TRANSFER RECONSTRUCTION

For one Fourier triad let modal energy transfers be

$$T=(T_1,T_2,T_3)$$

with exact energy conservation:

$$T_1+T_2+T_3=0$$

Let:

$$x_i=h|k_i|^2$$

$$p_a(x)=a\ x/(1+a\ x)$$

$$R_a(T)=\Sigma_i\ p_a(x_i)\ T_i$$

Define:

$$Z_h(T)=\Sigma_i\ x_i\ T_i$$

which is h times the triad enstrophy-transfer weight.

Let:

$$D_a=\Pi_i(1+a\ x_i)$$

Exact derived identity:

$$(D_a/a)\ R_a = Z_h - a\ W_h$$

for an auxiliary $W_h$.

Evaluate at a=1 and a=q:

$$Z_h = [\ D_q\ R_q\ /\ q - q\ D_1\ R_1\ ]/(1-q)$$

Therefore TWO clock channels reconstruct the full triad enstrophy coordinate exactly.

On a compact face:

$$h\ |\dot{Z}_\tau| \le C\ (|R_1|+|R_q|)$$

Thus critical radial enstrophy transfer cannot hide from the first two clock-factor probes.

## 17. GENERAL THREE-POINT SPECTRAL INTERPOLATION

For any $C^2$ spectral weight w on a compact face interval and any three radii $x_1,x_2,x_3$:

There exist bounded coefficients $A_\tau,B_\tau,C_\tau$ such that

$$w(x_i) = C_\tau + A_\tau\ p_1(x_i) + B_\tau\ p_q(x_i)$$

Energy conservation removes $C_\tau$:

$$\Sigma_i\ w(x_i)\ T_i = A_\tau\ R_1 + B_\tau\ R_q$$

The coefficients remain uniformly bounded even when radii collide.

Interpretation: On one three-mode interaction, two clock coordinates are local coordinates for every smooth spectral energy observable modulo constants.

## 18. EQUAL-RADIUS ANGULAR TRICHOTOMY

Exact equal-radius triad:

$$p+q=k$$

$$|p|=|q|=|k|=\kappa$$

Let n be the normal to the triad plane.

Write divergence-free input polarizations:

$$U_{hat}(p)=\alpha_p\ t_p + \beta_p\ n$$

$$U_{hat}(q)=\alpha_q\ t_q + \beta_q\ n$$

Define:

$$W_{pq} = \alpha_p\ \beta_q - \alpha_q\ \beta_p$$

Derived local exact formula:

$$E_k(p,q) = (\sqrt{3}/2)\ \kappa\ W_{pq}\ n$$

Thus:

$$W_{pq}=0 \iff \text{physical equal-radius Euler birth is zero.}$$

Angular response wedge:

$$k_{hat} \cdot (P_k\ U(q) \times P_k\ U(p)) = -\ W_{pq}\ /\ 2$$

Therefore loss of angular response rank equals physical birth loss, unless an ordered activity coefficient vanishes.

Local trichotomy:

A) $W_{pq}=0$:

Euler-null.

B) $W_{pq} \neq 0$ and $\alpha_p\ \alpha_q \neq 0$:

ordered responses span $k^\perp$;

no nonzero local cokernel witness survives.

C) W_pq != 0 and α_p α_q = 0:
    one input is exactly normal to the triad plane;
    this is the shear-channel geometry.
Hence:
    equal-radius active triad
      => Euler-null
         OR full tangent rank
         OR shear channel.
This is LOCAL and still needs independent audit before freezing.

## 19. BELTRAMI / SINGLE-HELICITY NULLITY

For a curl eigenspace:
    curl u = ±κ u
on one Stokes radius, standard vector identities give:
    B(u,v)+B(v,u)=0
for u,v in the same helicity eigenspace.
In particular:
    B(u,u)=0
Thus a Candidate-F symmetric covariance supported in one fixed single-helicity eigensphere is physically null.
This enlarges the known null geometries beyond simple shears.

## 20. Q1–Q2 ALGEBRAIC BRIDGE

Let:
    M_{σ,h}=m_σ(hA)
    N_{σ,h}=g(B^σ hA)
Frozen exact:
    ∂_σ M_{σ,h}
      = (log B) M_{σ,h} N_{σ,h}
For finite-Galerkin transport C:
    [C,M_{σ,h}] M_{σ,h}^{-1}
      = (log B)
        ∫_0^σ
        M_{τ,h}
        [C,N_{τ,h}]
        M_{τ,h}^{-1}
        dτ
At σ=1/2, the critical physical half-order commutator is algebraically generated by shell-number commutators.

### WARNING:

This is algebraic, not an automatic analytic norm equivalence. The conjugations change Sobolev order, and the physical Q1 transport differs from the regularized Q2 transport.

## 21. RESOLVENT DECOMPOSITION

Define:
    P_s = sA(I+sA)^(-1)
    R_s = (I+sA)^(-1)
Exact:
    N_h = Σ_{n>=0} P_{q^n h}
For any operator C:
    [P_s,C]
      = s R_s [A,C] R_s
up to sign convention.
Thus:
    [N_h,C]
      = Σ_{n>=0}
        q^n h
        R_{q^n h}
        [A,C]
        R_{q^n h}
Interpretation: The shell-number commutator is a logarithmic ladder of resolvent-sandwiched Stokes commutators.

## 22. CURRENT LEFT-PATH THEOREM TARGET

### OPEN THEOREM L:

Quantitative finite-face response theorem.
Prove that the remaining block
    |p|~|r|~|k|
    h|p|^2~h|r|^2~h|k|^2~1

cannot sustain a nonzero covariant response obstruction unless its dynamics lie in a physically null/depleted class.

A successful theorem should yield:

```
C_h -> 0
```

in a Candidate-F-compatible topology.

Then:

```
C_h -> 0
  => u_{qh}-u_h -> 0
  => G_K^disp -> 0
```

Possible null/depleted alternatives already identified: - radial conservation-nullity - equal-radius Euler-birth nullity - exact shear-null geometry - single-helicity/Beltrami nullity - possibly a quantitatively sparse exterior interaction class

## 23. WHAT IS STILL OPEN

The project is NOT a completed proof.

Major open items: 1. Full independent audit of the far-interaction X_T/Y_T parabolic theorem. 2. A complete Galerkin-uniform finite-face response theorem. 3. Profile extraction/compactness if face continuity fails. 4. Full treatment of the mollifier residual G_K^moll in the same certificate topology. 5. A concrete final a-posteriori NSE certificate with exact residual space and constants. 6. The residual-doubling theorem must be defined concretely and proved; do not assume it. 7. Full physical Q1 critical commutator theorem remains analytically distinct from Q2. 8. Terminal large-data critical flux inequality remains open.

## 24. EXPLICITLY FALSIFIED / DO NOT REUSE

Do NOT use these as proof inputs:

- ||M^{(K)}|| -> 0 in raw operator norm. FALSE.
- Candidate-F matrix is low rank. FALSE.
- Universal critical estimate ||u_{qh}-u_h|| <= C h^α. FALSE as a universal statement.
- Raw exponential locality of arbitrary NSE triads. FALSE. - Static polarization lock alone implies depletion. FALSE. - First Euler-order persistence alone implies depletion. FALSE. - Global B2 classification/globalization. FALSE. - Shear-nullity alone proves global regularity. NOT PROVED. - Ancient representation ladder exists automatically after Q2 failure. OPEN. - Full universal stress M_* is automatically the blow-up limit. OPEN. - Residual ideal-doubling theorem may not be treated as frozen without a concrete residual operator and proof.

## 25. SHORTEST CURRENT RESEARCH PROGRAM

A. Independently audit the derived covariant parabolic far theorem:

```
η_L ~ (1+L) 2^{-L/2}
```

B. Freeze the finite-width moving-face reduction if the audit survives.

C. Attack only:

```
hA ~ 1
comparable-frequency triads
```

D. Use:

```
clock-factor radial separation
two-clock transfer reconstruction
equal-radius angular trichotomy
shear/Beltrami null classes
```

E. Prove the finite-face response theorem:

```
C_h -> 0
```

or classify any surviving obstruction into a physically null/depleted/sparse class.

F. Then use frozen parallel transport:

```
C_h -> 0
=> u_{qh}-u_h -> 0
```

G. Then use no-rate Candidate F:

```
u_{qh}-u_h -> 0
=> G_K^disp -> 0
```

H. Close the mollifier residual and match one explicit a-posteriori certificate.

## 26. CURRENT CENTRAL IDEA IN ONE LINE

The LEFT path has compressed the scale-response problem from an unrestricted h->0 nonlinear PDE question to a finite-width moving-face problem:

```
far triads are summably perturbative
+ comparable off-face triads are clock-suppressed
+ radial cancellation is detectable by clock tangents
+ exact equal-radius angular degeneracy is Euler-null/full-rank/shear
```

so the remaining hard core is:

```
a Galerkin-uniform quantitative theorem for the comparable-frequency block at hA ~ 1.
```

## END OF CONTEXT

# Appendix B - Critical Context Transfer Ledger

Raw project-context snapshot appended for provenance and reuse. It may contain duplicated material, research-status labels, and open/falsified branches; it is not promoted by inclusion into the Level-0 theorem.

**BKQR / 3D NAVIER–STOKES**

**CRITICAL CONTEXT TRANSFER FOR A NEW PROMPT**

Internal research summary — 14 August 2026

**USE THIS FILE AS CONTEXT IN A NEW CHAT.**

It is a compact transfer of the current mathematical state of the BKQR/Navier–Stokes project. It is NOT a claim that 3D Navier–Stokes global regularity has been proved.

**0. STATUS DISCIPLINE**

The project uses four levels of confidence:

**1. FROZEN / EXACT**

Algebraic identities or finite-Galerkin / bounded-functional-calculus facts that have survived audit.

**2. DERIVED / AUDITED**

Nontrivial deductions with a proof sketch or finite-dimensional proof, but which may still need publication-level domain/topology formalization.

**3. STANDARD-NEIGHBOR**

Classical PDE / functional-analysis input expected to be standard once specialized precisely.

**4. OPEN**

Genuine missing theorems, possibly carrying the hard 3D Navier–Stokes difficulty.

Important: do not silently promote an OPEN statement to a theorem. The program is currently a proof architecture with a large exact algebraic core and a small number of serious analytic closure obligations.

Primary working domain:

    Omega = T^3,

    divergence-free, mean-zero vector fields,

    viscosity nu > 0,

    A = -P Delta >= 0  (Stokes operator),

    B(u,v) = P div(u tensor v).

Physical time is t. Representation / regularization scale is h > 0. Geometric ratio is q in (0,1), often q = 1/2. K is annihilation / correction depth.

In 3D:

    u(x,t) in R^3,

    u tensor v in R^{3x3},

    div(u tensor v) in R^3,

    P projects to divergence-free output.

For a Fourier mode k, incompressibility restricts the polarization to k^\perp, a 2D plane.

**1. STABLE GEOMETRIC ANNIHILATOR**

Define the scale shift:

    (S F)(h) = F(q h).

For K >= 1 define:

    P_K(S) = product_{m=1}^K (S - q^m I)/(1 - q^m)

           = sum_{j=0}^K c_j^(K) S^j.

Then exactly:

    sum_j c_j^(K) = 1,

    sum_j c_j^(K) q^{jn} = 0,   1 <= n <= K.

Equivalently:

    P_K 1 = 1,

    P_K h^n = 0,   1 <= n <= K.

The weights are the Lagrange interpolation coefficients for evaluation at h=0 from the geometric nodes

    h, qh, q^2 h, ..., q^K h.

Closed form:

    c_j^(K)

    =

    (-1)^(K-j) q^{(K-j)(K-j+1)/2}

    /

    [ (q;q)_j (q;q)_{K-j} ].

Most important stability fact:

```
    sum_{j=0}^K |c_j^(K)|
    =
    product_{m=1}^K (1+q^m)/(1-q^m)
    <= C_q < infinity
```

for fixed q in (0,1), uniformly in K.

Interpretation: K can become arbitrarily large without the coefficient norm blowing up, provided q is fixed away from 1.

## 2. CORRECTION LAYERS, SCALE LADDER, AND EXACT q-PRODUCT RULE

Define:

```
    C_0 = I,
    C_K = P_K - P_{K-1},   K >= 1.
```

Then:

```
    C_K
    =
    product_{m=0}^{K-1}(S-q^m)
    /
    product_{m=1}^K(1-q^m).
```

Exact scale-ladder relation:

```
    C_K F(qh)
    =
    q^K C_K F(h)
    +
    (1-q^{K+1}) C_{K+1} F(h).
```

This implies descent: if a fixed higher correction layer C_{K+1}F tends to zero along h_r=h_0 q^r, then C_K F tends to zero, and iterating eventually gives

```
    F(qh)-F(h) -> 0.
```

This is the basis of the higher-scale-curvature route to neighboring-scale continuity.

Exact bilinear scale-lift:

For every h-independent bilinear map Q:Y1 x Y2 -> X,

```
    C_K Q(F,G)(h)
    =
    sum_{r=0}^K
    Q(
      C_r F(q^{K-r}h),
      C_{K-r}G(h)
    ).
```

Define the q-convolution:

```
    (V star_q^Q W)_K(h)
    =
    sum_{r=0}^K
    Q(
      V_r(q^{K-r}h),
      W_{K-r}(h)
    ).
```

Then:

```
    C[Q(F,G)] = C[F] star_q^Q C[G].
```

For multiplication this gives an exact multiplicative scale algebra. For Navier–Stokes it gives:

```
    C_K B(F,G)
    =
    sum_{r=0}^K
    B(
      C_r F(q^{K-r}h),
      C_{K-r}G(h)
    ).
```

This is one of the strongest abstract BKQR theorems: the quadratic NSE nonlinearity becomes an exact q-convolution in representation grade.

## 3. COMPATIBLE TOWERS AND MULTIPLICATION-AWARE BANACH ALGEBRA

A tower V=(V_0,V_1,...) is compatible when:

```
    V_K(qh)
    =
    q^K V_K(h)
    +
    (1-q^{K+1}) V_{K+1}(h).
```

The map

```
    F -> C[F] = (C_0F, C_1F, C_2F, ...)
```

is a bijection from ordinary scale families to compatible towers. The inverse is simply F = V_0.

So the BKQR tower representation is lossless. The many-to-one step only occurs when one later projects/evaluates at the physical boundary h=0.

If the base bilinear map satisfies:

    ||Q(a,b)||_X <= M_Q ||a||_{Y1} ||b||_{Y2},

and weights w_K are submultiplicative:

    w_{r+s} <= w_r w_s,

then with the weighted l^1 tower norm,

    ||V star_q^Q W||_{X,w}
    <=
    M_Q ||V||_{Y1,w} ||W||_{Y2,w}.

Thus compatible towers form a Banach algebra whenever the base physical space is a Banach algebra.

Boundary-vanishing ideals also multiply: if

    I_mu = {F : ||F(h)|| <= C mu(h), mu(h)->0},

then for bounded bilinear Q,

    Q(I_mu, I_nu) subset I_{mu nu}.

In particular:

    Q(I_mu, I_mu) subset I_{mu^2}.

This is important because quadratic improvement does NOT require a power law. Examples:

    mu(h)=h^alpha        -> mu^2=h^{2alpha},
    mu(h)=1/log(1/h)     -> mu^2=1/log^2(1/h).

## 4. POSITIVE BKQR CLOCK

Let E_r ~ Exp(1) independently and define:

    Y_infinity = sum_{r>=0} q^r E_r.

Its Laplace transform is:

    f_q(z)
    =
    E exp(-z Y_infinity)
    =
    product_{r>=0} (1+q^r z)^(-1).

The self-similarity identity is exact:

    f_q(qz) = (1+z) f_q(z).

For the Stokes operator define:

    J_h = f_q(hA).

Then:

    J_{qh} = (I+hA) J_h,

and therefore:

    J_{qh} - J_h = h A J_h.

Also:

    J_h = E[ exp(-h Y_infinity A) ],

so J_h is an average of positive Stokes semigroups.

The clock identity and the geometric annihilator are therefore governed by the same q-scale structure.

## 5. POSITIVE q-SHELL PARTITION

Define:

    Pi_K(hA) = C_K J_h.

At scalar spectral variable x>=0:

    pi_K(x)
    =
    q^{K(K-1)/2}
    x^K f_q(x)
    /
    (q;q)_K.

Then:

    pi_K(x) >= 0,
    sum_{K>=0} pi_K(x) = 1.

Hence:

    Pi_K(hA) >= 0,
    sum_K Pi_K(hA) = I.

Exact raising law:

    x pi_K(x)
    =
    (q^{-K}-q) pi_{K+1}(x),

so:

    h A Pi_K(hA)

```
    =
    (q^{-K}-q) Pi_{K+1}(hA).
Thus K is a positive logarithmic spectral coordinate rather than an arbitrary correction index.
```

## 6. BERNOULLI SHELL LAW AND LOGARITHMIC LOCALIZATION

```
The shell generating function is:
    sum_K t^K pi_K(x)
    =
    f_q(x)/f_q(tx)
    =
    product_{r>=0}
    (1+t q^r x)/(1+q^r x).
Define independent Bernoulli variables:
    xi_r(x) ~ Bernoulli( p_r(x) ),
    p_r(x) = q^r x/(1+q^r x).
Then:
    N_x = sum_r xi_r(x),
    P(N_x=K)=pi_K(x).
Mean shell number:
    g(x)=E N_x
        = sum_r q^r x/(1+q^r x).
Exact logarithmic derivative:
    g(x) = -x d/dx log f_q(x).
Exact Fisher/variance identity:
    x g'(x) = Var(N_x).
The variance is uniformly bounded for fixed q. Also:
    g(x) = log_B x + O_B(1),
    B=q^{-1}.
Thus the shell index is localized near:
    K ~ log_B(h lambda).
```

## 7. SHELL MOMENTS RECOVER SOBOLEV ORDER

```
Define:
    m_sigma(x)
    =
    sum_K B^{sigma K} pi_K(x)
    =
    f_q(x)/f_q(B^sigma x).
For 0 <= sigma <= 1:
    m_sigma(x) ~_{B,sigma} (1+x)^sigma.
Therefore:
    M_sigma(u)
    :=
    <u, m_sigma(hA) u>
    ~
    ||(I+hA)^{sigma/2} u||_2^2.
At the critical index sigma=1/2:
    M_{1/2}(u)
    ~
    ||u||_2^2
    +
    h^{1/2} ||A^{1/4}u||_2^2.
On T^3:
    H^{1/2} embeds into L^3.
So bounded critical shell moment controls the standard critical L^3 scale.
```

## 8. EXACT NAVIER–STOKES SHELL DYNAMICS

```
For a smooth NSE solution:
    partial_t u + nu A u + B(u,u) = 0,
define:
    E_K = <u, Pi_K u>,
    D_K = <A^{1/2}u, Pi_K A^{1/2}u>,
    T_K = -<B(u,u), Pi_K u>.
Then exactly:
    (1/2) partial_t E_K + nu D_K = T_K.
Because sum_K Pi_K = I and convection is energy-skew:
```

```
    sum_K T_K = 0.
The raising law gives:
    D_K = (q^{-K}-q)/h * E_{K+1}.
Weighted shell transfer: with W_sigma=m_sigma(hA),
    T_sigma
    =
    -<B(u,u), W_sigma u>
    =
    -(1/2)<[W_sigma, u·grad]u, u>.
At sigma=1/2:
    T_{1/2}
    =
    -(1/2)
    <[m_{1/2}(hA), u·grad]u, u>.
Interpretation: critical 3D transfer is exactly a failure of transport to commute with the critical scale measurement.
```

## 9. POSITIVE REGULARIZED NSE FAMILY AND EXACT RESIDUAL

```
For each h>0 use:
    partial_t u_h
    + nu A u_h
    + B(J_h u_h, u_h)
    = 0,
with the same initial datum.
Sample:
    h_j = h q^j,
    u_j = u_{h_j}.
Define:
    U_K = sum_{j=0}^K c_j u_j,
    d_j = u_j - U_K.
Because sum_j c_j = 1:
    sum_j c_j d_j = 0.
Then U_K satisfies the unregularized NSE with an exact stress:
    partial_t U_K
    + nu A U_K
    + B(U_K,U_K)
    =
    P div G_K,
where:
    G_K
    =
    - sum_j c_j d_j tensor d_j
    +
    sum_j c_j [(I-J_j)u_j] tensor u_j.
Write:
    G_K = G_K^disp + G_K^moll.
Most important structural point: all first-order cross-scale disagreement cancels exactly. The surviving
solution-family dispersion begins quadratically:
    G_K^disp = -sum_j c_j d_j tensor d_j.
This is a special NSE instance of the general affine quadratic-covariance identity.
```

## 10. CANDIDATE F GREEN KERNEL

```
Define neighboring representation increments:
    delta_r = u_{r+1} - u_r.
Define:
    tau_r = sum_{j=r+1}^K c_j,
    p_r   = sum_{j=0}^r c_j = 1-tau_r.
Then:
    G_K^disp
    =
    - sum_{r,s<K}
      M_rs^(K)
      delta_r tensor delta_s,
with exact Green kernel:
    M_rs^(K)
    =
    p_{min(r,s)} tau_{max(r,s)}.
```

The inverse is tridiagonal:
    (M^(K))^{-1}
    =
    D_K^*
    diag(c_0^{-1},...,c_K^{-1})
    D_K.
Interpretation: M^(K) is the Green matrix of a signed one-dimensional scale Laplacian.
Important falsification:
    ||M^(K)|| -> 0
is FALSE. The matrix is full rank, indefinite, and its finest-corner entries remain O(1).
What survives is localization:
    |M_rs^(K)|
    <=
    C_q
    q^{(K-min(r,s))(K-min(r,s)+1)/2}.
For D_rho=diag(1,rho,rho^2,...), 0<rho<1:
    ||D_rho M^(K) D_rho||
    <=
    C(q,rho) rho^{2K}.
Even stronger: if
    sup_r ||delta_r||_Y < infinity
and
    delta_r -> 0 in Y,
then, for any bounded tensor map Y x Y -> X,
    ||G_K^disp||_X -> 0.
No universal h^alpha rate is required. Mere tail continuity can collapse the dispersion residual.

## 11. UNIVERSAL CO-MOVING CORNER

Reverse indices:
    a = K-1-r,
    b = K-1-s.
Then the finest-scale coefficients converge:
    c_{K-m}^(K)
    ->
    C_m
    =
    (-1)^m q^{m(m+1)/2}
    /
    [ (q;q)_infinity (q;q)_m ].
Set:
    T_a = sum_{m=0}^a C_m.
The co-moving Green matrix converges to:
    M_star,ab
    =
    T_{min(a,b)}
    [1 - T_{max(a,b)}].
The convergence is in entrywise l^1, hence trace/nuclear norm and operator norm.
Therefore:
    ||M^(K)||_op -> ||M_star||_op > 0.
Interpretation: the obstruction does not disappear. It travels toward the finest representation scale and converges to a universal q-dependent profile.
Signed covariance representation: with
    H_ma = 1_{m<=a} - T_a,
one has:
    M_star,ab = sum_m C_m H_ma H_mb.
For scale increments Delta_a and tails
    Y_m = sum_{a>=m} Delta_a,
    Y_bar = sum_m C_m Y_m,
the quadratic form becomes:
    sum_{a,b} M_star,ab Delta_a tensor Delta_b
    =
    sum_m C_m (Y_m-Y_bar) tensor (Y_m-Y_bar).
So the universal Candidate-F corner is a signed covariance of the representation-scale path.

## 12. SHELL-NUMBER OPERATOR AND COVARIANT SCALE TANGENT

Define the mean shell-number operator:

```
    N_h = g(hA).
Define:
    K_h = -h partial_h J_h.
Exact factorization:
    K_h = J_h N_h.
For divergence-free transport C_a v = P(a·grad v), at finite Galerkin level:
    [C_a,N_h]
    =
    sum_{K,J}
    (J-K) Pi_K C_a Pi_J.
Thus the shell-number commutator measures the first shell-distance moment of nonlinear transfer.
Fourier symbol: for k=p+r, the commutator carries the factor
    g(h|r|^2)-g(h|k|^2).
Two key exact/derived consequences:
1. Equal-radius nullity:
       |r|=|k|  =>  contribution = 0.
2. Low-high derivative transfer: if |p| <= |r|/2,
       |r| |g(h|r|^2)-g(h|k|^2)|
       <= C_q |p|.
So the commutator moves the derivative off the high transported mode and onto the low advecting mode.
Now define:
    Z_h = -h partial_h u_h.
Let L_h be the linearization:
    L_h v
    =
    partial_t v
    + nu A v
    + B(J_h v,u_h)
    + B(J_h u_h,v).
Differentiating the regularized NSE:
    L_h Z_h = -B(K_h u_h,u_h).
Define the covariant scale tangent:
    C_h = Z_h + N_h u_h.
Then the deterministic clock transport cancels exactly and:
    L_h C_h
    =
    [ C_{J_h u_h}, N_h ] u_h.
This is a central exact identity: representation-scale motion, after subtracting the clock connection, is driven by
the shell-number commutator.
```

## 13. Q1–Q2 ALGEBRAIC BRIDGE

```
Let:
    M_{sigma,h}=m_sigma(hA),
    N_{sigma,h}=g(B^sigma hA).
Then:
    partial_sigma M_{sigma,h}
    =
    (log B) M_{sigma,h} N_{sigma,h}.
For a fixed transport operator C:
    [C,M_{sigma,h}] M_{sigma,h}^{-1}
    =
    (log B)
    integral_0^sigma
    M_{tau,h}
    [C,N_{tau,h}]
    M_{tau,h}^{-1}
    d tau.
At sigma=1/2, the critical physical commutator is generated algebraically by the same shell-number commutators that
drive the covariant scale tangent.
Interpretation: Q1 (critical physical transfer control) and Q2 (representation-scale continuity) are not unrelated
problems. They are two projections of the same scale-commutator geometry.
```

## 14. ENERGY-SPHERE GEOMETRY

```
For each physical time t, define the trajectory vector:
    Ucal_h(t)
    =
    (
```

```
      u_h(t),
      sqrt(2 nu) A^{1/2}u_h|_[0,t]
    ).
The fixed-h energy identity gives:
    ||Ucal_h(t)|| = ||u_0||_2
for every h.
Thus the entire representation-scale family lies on one Hilbert sphere.
Three-scale curvature identity: Let
    X=Ucal_h,
    Y=Ucal_{qh},
    Z=Ucal_{q^2h},
all with norm R.
Set:
    Delta_h   = Y-X,
    Delta_qh  = Z-Y,
    E_{alpha,q}
    =
    Z-(1+q^alpha)Y+q^alpha X
    =
    Delta_qh - q^alpha Delta_h.
Then exactly:
    <Y,E_{alpha,q}>
    =
    -(1/2)
    (
      ||Delta_qh||^2
      +
      q^alpha ||Delta_h||^2
    ).
Hence:
    ||Delta_h||^2
    <=
    (2R/q^alpha)
    ||E_{alpha,q}||.
This is the square-root curvature descent: small second scale curvature forces even smaller first scale displacement.
Continuous form: with s=-log h and X(s)=Ucal_{e^{-s}},
    ||X(s)||=R,
and for
    C_alpha = X_ss + alpha X_s,
one has:
    <X,C_alpha> = -||X_s||^2,
so:
    ||X_s||^2 <= R ||C_alpha||.
If:
    integral ||C_alpha(s)||^{1/2} ds < infinity,
then the representation-scale trajectory has finite path length and converges strongly.
```

## 15. ENERGY-LEVEL CLOCK FORCING GAIN

```
The clock derivative and second clock curvature satisfy scalar multiplier bounds leading to:
    ||K_h v||_2
    +
    ||H_h v||_2
    <=
    C_q h^{1/2} ||A^{1/2}v||_2.
For the energy-controlled regularized family:
    ||K_h u_h||_{L_t^2 L_x^2}
    +
    ||H_h u_h||_{L_t^2 L_x^2}
    <=
    C h^{1/2}.
Using u_h in L_t^2 L_x^6:
    ||(K_h u_h) tensor u_h||_{L_t^1 L_x^{3/2}}
    <=
    C h^{1/2}.
Thus the scale-forcing already has a genuine positive h^{1/2} gain at energy level.
```

However: L_t^1 L_x^{3/2} is borderline for direct propagation to L_t^\infty L_x^3, because the Stokes kernel behaves like (t-s)^{-1}. So this estimate alone does not close the problem.

## 16. HP ADIABATIC RESPONSE DICHOTOMY — OPEN HIGH PRIORITY

The tangent equation is:

L_h Z_h
=
-P div F_h,
F_h = (K_h u_h) tensor u_h.

Baseline:

||F_h|| <= C h^{1/2}.

Desired positive-response alternative: prove in compatible spaces that

||L_h^{-1}||
<=
C h^{-theta_T},
with theta_T < 1/2.

Then:

||Z_h||
<=
C h^{alpha_T},
alpha_T = 1/2 - theta_T > 0,

and therefore:

||u_{qh}-u_h||
<=
C h^{alpha_T}.

Candidate F then squares this:

G_K^disp gains roughly h^{2 alpha_T}.

If such an inverse estimate fails, the failure should yield an adjoint quasimode / resonance. The current program then asks Q1 locality to force that resonance into the moving physical face hA ~ 1 and classify it geometrically.

This is OPEN and one of the main present analytic targets.

## 17. EXACT SHEAR-NULL ALGEBRAS

For a nonzero direction n define:

N_n
=
{ W(x)=F(n·x) : F·n=0 }.

Then for all W1,W2 in N_n:

B(W1,W2)=0.

Likewise for fixed nonzero real v:

N_v
=
{ v phi(x) : v·grad phi = 0 }.

Then:

B(W1,W2)=0

for all W1,W2 in N_v.

These null algebras are preserved by:

A,
f(A),
J_h,
I-J_h.

So if an entire representation ladder lies in one common shear-null algebra, every internal bilinear interaction vanishes exactly.

## 18. UNIVERSAL-CORNER SHEAR-NULL THEOREM

Suppose the co-moving representation ladder U^(a) lies in one fixed shear-null algebra N for all a.

Then every increment:

Delta_a = U^(a)-U^(a+1)

also lies in N.

Therefore:

P div(Delta_a tensor Delta_b)=0

for every pair.

Also:

(I-J^(a))U^(a) in N,

so the mollifier-defect tensors are null as well.

Hence the full universal Candidate-F stress satisfies:

P div G_star = 0.

Key replacement principle:

do NOT require M^(K) -> 0.

The matrix can stay order one. What matters is whether the input seen by the universal corner lies in a physical null geometry.

## 19. FOURIER-LINE TRIANGULARITY AND LOCAL C4 DICHOTOMY

Let Pi_n project onto the Fourier line {m n : m in Z}. Write:

    U = W+V,

    W = Pi_n U.

Then W lies in the shear algebra N_n and:

    Pi_n B(W,W)=0,

    Pi_n B(W,V)=0,

    Pi_n B(V,W)=0.

Thus:

    partial_t W + nu A W + Pi_n B(V,V)=0.

Only exterior-exterior interactions:

    V+V -> W

can inject into the shear line.

Local degenerate-C4 theorem: If two distinct rank-defective output planes share one nonzero polarization e, then any persistent common newborn response must either:

1. provide a second independent shared response, forcing plane synchronization; or

2. remain collinear with e.

In the second case incompressibility forces every common birth wavevector q to satisfy:

    q·e=0.

The resulting common-birth field is an exact shear-null field and:

    B(W,W)=0,

    (omega_W·grad)W=0.

Therefore locally:

    persistent degenerate C4

    =>

    plane synchronization

    OR

    exact common-sector shear nullity.

This is a LOCAL theorem only. No global decomposition of every possible obstruction has been proved.

## 20. EXACT NONLINEAR ERROR-ARROW / RESIDUAL RECURSION

For one affine correction:

    T_a = (S-aI)/(1-a),

    U^+ = T_a U,

    C = U^+ - U.

For a nonlinear map N define the nonlinear commutator:

    C_Ta[N](U)

    =

    N(T_a U)-T_a N(U).

The Taylor expansion has no zeroth- or first-order term:

    C_Ta[N](U)

    =

    sum_{r>=2}

    [1-(1-a)^{r-1}]/r!

    D^rN(U)[C^{tensor r}].

For a quadratic map N(U)=Q(U,U):

    C_Ta[N](U)

    =

    a Q(C,C).

Thus the newly created nonlinear defect is exactly quadratic.

For successive corrections:

    U_m=P_m U_0,

    P_m=T_m...T_1,

    C_m=U_m-U_{m-1},

and residual:

    G_m=N(U_m)-P_m N(U_0),

one gets the exact recursion:

    G_m

    =

    T_m G_{m-1}

    +

    q^m Q(C_m,C_m).

So every step contains:

transported old residual

+

new quadratic defect.

## 21. EXACT BANDWISE VALUATION DOUBLING

This is the correct exact algebraic version of:

R_K(I^[n]) subset I^[2n].

In an integer-graded analytic/formal scale algebra define:

I_n

=

{ F(h)=sum_{k>=n} a_k h^k }.

For each grade r define:

T_r=(S-q^r I)/(1-q^r).

Then:

T_r(h^k)

=

(q^k-q^r)/(1-q^r) h^k,

so:

T_r(h^r)=0.

Define the complete band annihilator:

R_n

=

product_{r=n}^{2n-1} T_r.

It kills every grade:

n,n+1,...,2n-1.

Therefore exactly:

R_n(I_n) subset I_{2n}.

Uniform stability:

||R_n||

<=

product_{r=n}^{2n-1}

(1+q^r)/(1-q^r)

<= C_q.

Dyadic tower:

I_1

-> I_2

-> I_4

-> I_8

-> ...

using:

R_1, R_2, R_4, R_8, ...

and:

R_{2^{m-1}} ... R_2 R_1

=

P_{2^m-1}.

Thus the original geometric annihilator already contains an exact dyadic doubling tower when its factors are grouped into bands.

## IMPORTANT STATUS DISTINCTION:

## EXACT:

R_n(I_n) subset I_{2n}

in the declared graded analytic/formal algebra.

## STILL OPEN:

construct one concrete Navier–Stokes residual Banach space X_T and one concrete full residual-correction operator R_K :X_T->X_T such that the actual NSE residual satisfies the same ideal-doubling structure together with:

||R_K X||_{X_T}

<=

C_T ||X||_{X_T}^2.

Do not conflate these two statements.

## 22. CURRENT PROOF CHAIN

The current compressed proof architecture is:

stable geometric annihilator

->

exact correction-layer / q-convolution algebra

```
    ->
positive BKQR clock
    ->
positive logarithmic shell calculus
    ->
globally controlled fixed-h regularized family
    ->
exact quadratic residual G_K
    ->
control G_K^disp via:
    Candidate-F tail continuity
    / covariant commutator
    / energy-sphere curvature
    / obstruction nullification
    ->
control G_K^moll in the same residual topology
    ->
certificate-compatible small seed
    ->
concrete NSE residual doubling if needed
    ->
a-posteriori NSE certificate
    ->
regularity on [0,T]
    ->
global regularity if this works for every finite T.
```

The final deduction is short. The difficult work lies in producing the non-circular small seed and matching everything in one certificate-compatible topology.

## 23. CURRENT MAIN OPEN TARGETS

### Q1. FULL THREE-FREQUENCY COMMUTATOR THEOREM

Prove a Galerkin-uniform critical-space theorem for the shell-number / moment commutator.

Equal-radius nullity and low-high derivative transfer are already known.

The hard missing region is especially high-high -> low.

### Q2. PHYSICAL-FACE / NEIGHBORING-SCALE CONTINUITY

Prove u_{qh}-u_h -> 0 in a useful non-circular topology,

or prove decay of a higher correction layer and descend,

or prove adiabatic curvature decay and use the energy-sphere theorem.

### Q3R. SCALE-RESPONSE RESONANCE

If the inverse linearized response loses >= h^{-1/2}, extract the adjoint quasimode and classify it.

### Q3F. FACE CONCENTRATION / PROFILE RENEWAL

If Q2 fails by concentration rather than response resonance, establish profile extraction / compactness at the mo ving face.

### SMALL SEED

Find one concrete residual space X_T and prove:

$$C_T ||G_K||_{X_T} < 1$$

for finite h,K without using a hidden critical regularity assumption.

### CONCRETE NSE DOUBLING

Define the actual residual correction operator in the same X_T and prove the norm-quadratic squaring theorem.

### A-POSTERIORI CERTIFICATE

Freeze one exact downstream theorem, one norm X_T, and all constants.

### PARALLEL SHELL ROUTE TERMINAL TARGET

A large-data critical flux inequality of the form:

$$|T_{1/2}| <= theta nu D_{1/2} + C(1+M_1)M_{1/2},$$

$$theta<1.$$

This would close the shell route, but it is currently OPEN.

## 24. FALSIFICATION ARCHIVE — DO NOT REUSE

The following statements have been tested and must NOT be used as proof inputs:

1. Raw matrix contraction:

||M^(K)|| -> 0.

**FALSE.**

Correct replacement: universal moving corner M_star != 0.

2. Low-rank Candidate-F matrix.

**FALSE.**

M^(K) is full rank.

3. Universal critical operator power law:

||(J_{qh}-J_h)v||_critical <= C h^alpha ||v||_critical.

FALSE at the moving face hA~1. This does not rule out PDE-structured scale continuity of u_h.

4. Raw exponential locality of arbitrary NSE triads.

**FALSE.**

The useful gain appears after energy cancellation produces a commutator.

5. Static polarization lock => nonlinear depletion.

FALSE in general.

6. First Euler-order persistence => depletion.

FALSE; finite Fourier counterexamples exist in the project ledger.

7. Global B2 classification. FALSE / discarded. Only local robust/degenerate/sparse mechanisms that have independent proofs may be used.

8. Exact shear-nullity by itself proves global regularity.

**NOT PROVED.**

What is proved is self-nullity, triangular support, and energy-controlled shear forcing.

9. Every Q2 failure automatically produces an ancient representation ladder. OPEN; requires concentration compactness/profile extraction.

10. The full universal residual G_star is automatically a blow-up limit.

OPEN; requires actual convergence of the representation ladder and residual sums in a certificate topology.

## 25. WHAT IS MOST IMPORTANT TO PRESERVE IN A NEW CHAT

If continuing this research in a new prompt, do the following:

- Treat the exact algebraic layer as already available. - Do not spend time reproving elementary geometric annihilator identities unless auditing them. - Do not revive falsified routes. - Distinguish representation scale h from physical time t and from spatial Fourier scale. - Keep the Candidate-F matrix and the physical action P div separate: an order-one stress can be physically null. - Prefer theorem statements that are Galerkin-uniform and survive passage to the infinite-dimensional problem. - Avoid circularity: a proposed Q2 estimate is not useful if it already assumes a critical norm known to imply regularity. - Keep all upstream estimates in one eventual residual topology X_T. - The most plausible common lever is the three-frequency commutator theorem because it appears simultaneously in:

    critical physical transfer,

    covariant representation transport,

    shear-boundary terms,

    and localization of possible adjoint resonances.

- The other high-priority lever is the adiabatic response dichotomy:

    either the inverse response loses less than h^{-1/2}, giving positive scale decay,

    or its failure produces a structured obstruction to classify.

A good next research question is:

> "Can we prove a Galerkin-uniform three-frequency estimate for
> [C_a,N_h] (or its critical moment version)
> that handles high-high -> low interactions without assuming an a priori
> critical norm bound, and does the same estimate propagate covariant
> scale continuity strongly enough to make G_K small in a certificate
> topology?"

Alternative high-priority question:

> "Can we prove the HP adiabatic response dichotomy in a concrete parabolic
> pair Y_T, X_T, and show that every failure mode is forced either into a
> shear-null/synchronized geometry or into a quantitatively controlled
> exterior obstruction?"

## 26. SHORTEST CURRENT MATHEMATICAL SUMMARY

The project now has a large exact backbone:

    geometric extrapolation

    + exact q-product / bilinear scale-lift

    + multiplication-aware tower algebra

    + positive self-similar Stokes clock

    + positive logarithmic shells

+ exact NSE shell balances
+ exact critical commutator identity
+ exact quadratic solution-family residual
+ Candidate-F Green kernel
+ universal moving corner
+ covariant shell-number tangent
+ energy-sphere curvature descent
+ exact shear-null algebras
+ exact formal bandwise valuation doubling.

The current 3D Navier–Stokes proof boundary is not the algebra. It is the analytic step that must turn this structure into a non-circular, Galerkin-uniform, certificate-compatible small residual.

Do not claim the Millennium problem is solved unless those open analytic bridges are actually proved.

## SOURCE BASIS / INTERNAL PROVENANCE

This transfer summary is distilled from the project PDFs and theorem packages, especially:

- BKQR: A Calculus of Structured Finite Approximants - Continuous BKQR Embedding for Singular PDE Analysis - BKQR Novel Core Formula Handbook - BKQR to Navier–Stokes: Complex-Scale Annihilation, Continuum Completion, and a Multi-Scale Regularity Program - BKQR Continuous Compiler and Navier–Stokes Novel-Core Workbook - BKQR Candidate F research dossier - BKQR Mathematical Core Reference - BKQR / Navier–Stokes Frozen q-Scale Core (13 Aug 2026) - BKQR / 3D Navier–Stokes Frozen Theorem Core v2 (13 Aug 2026) - BKQR Audited Frozen Core (13 Aug 2026) - BKQR Universal-Corner and Shear-Null Theorems (13 Aug 2026) - BKQR Scale Algebra and the Current Navier–Stokes Core (13 Aug 2026) - BKQR Adiabatic Scale-Response for 3D Navier–Stokes (13 Aug 2026) - the current conversation's later bandwise-doubling audit.

## END OF CONTEXT TRANSFER

# Appendix C - Navier-Stokes 3D Framework Snapshot

Raw project-context snapshot appended for provenance and reuse. It may contain duplicated material, research-status labels, and open/falsified branches; it is not promoted by inclusion into the Level-0 theorem.

Yes. I would rebuild the whole object from the ground up and deliberately separate three things:

**(i)** the mathematical spaces required by 3D Navier–Stokes, **(ii)** the BKQR machinery that exists abstractly on those spaces before using anything special about Navier–Stokes, and **(iii)** several distinct proof directions that become available only after the two structures are coupled.

The uploaded sources support a common frozen core—geometric annihilation, positive clock/shells, the exact quadratic Candidate-F residual, and covariant scale transport—while the later handoffs develop different possible endgames: front/radial response, anisotropic profiles, shell flux/helicity, costed carrier geometry, holonomy, and quotient-line propagation. filecite turn0file0 filecite turn0file1 filecite turn0file4

# I. Bottom layer: metric spaces

Begin with only a metric space

$$(X,d).$$

There is no addition, scalar multiplication, inner product, Fourier transform, or PDE yet.

What can BKQR already see here?

Introduce a positive representation scale

$$h>0,$$

and a geometric ratio

$$0<q<1.$$

A representation family is simply a map

$$F:(0,h_0]\longrightarrow X.$$

Define the geometric refinement operation

$$(SF)(h):=F(qh).$$

At the pure metric level we can discuss only things such as

$$d(F(qh),F(h)),$$

Cauchy behavior as $h\to0$,

$$F(h_n)\to F_*$$

and neighboring-scale continuity

$$\boxed{ d(F(qh),F(h))\to0. }$$

So the **first primitive BKQR idea does not require linearity**:

> study an object by following its orbit along a geometric representation scale.

This level already distinguishes physical variables from representation variables. For Navier–Stokes, $t$ will be physical time; $h$ is not time.

---

# II. Add linear structure: normed and Banach spaces

Now suppose

$$(Y,\|\cdot\|_Y)$$

is a normed vector space, preferably Banach.

Then affine combinations of representation samples make sense.

For

$$F:(0,h_0]\to Y$$

the BKQR geometric annihilator is

$$\boxed{$$

**P_K(S)**

$$= \prod_{m=1}^K \frac{S-q^mI}{1-q^m} = \sum_{j=0}^K c_j^{(K)}S^j . }$$

Hence

$$P_KF(h) = \sum_{j=0}^K c_j^{(K)}F(q^jh).$$

The exact moment identities are

$$\sum_{j=0}^K c_j^{(K)}=1,$$

and

$$\sum_{j=0}^K c_j^{(K)}q^{jn}=0, \qquad 1\le n\le K.$$

Thus $P_K$ preserves constants but annihilates the first $K$ geometric powers.

Most importantly,

$$\boxed{ \sup_K\sum_{j=0}^K|c_j^{(K)}|<\infty }$$

for fixed $q\in(0,1)$.

So BKQR is not merely a formal extrapolator: it is **uniformly stable in correction depth**. filecite turn0file1

Define correction layers

$$[$$

**C_0=I,**

\qquad

**C_K=P_K-P_{K-1}.**

\]

Then

\[

**C_K**

= \frac{\prod_{m=0}^{K-1}(S-q^mI)} {\prod_{m=1}^{K}(1-q^m)} \]

and the tower satisfies

\[ \boxed{ C_KF(qh) = q^K C_KF(h) + (1-q^{K+1})C_{K+1}F(h). } \]

This already gives us a very useful abstract structure:

\[

**F**

\longmapsto \mathcal C(F) = (C_0F,C_1F,C_2F,\ldots). \]

The compatible tower is lossless: \(F\) is recovered from its zeroth component.

So BKQR has turned one function of \(h\) into a **scale jet**.

---

# III. Add bilinear structure: the multiplication-aware Banach layer

Now suppose we have Banach spaces

\[ Y_1,\qquad Y_2,\qquad X \]

and an \(h\)-independent bounded bilinear map

\[ Q:Y_1\times Y_2\to X. \]

Then a remarkable exact identity appears:

\[ \boxed{ C_K Q(F,G)(h) = \sum_{r=0}^K Q\!\left( C_rF(q^{K-r}h), C_{K-r}G(h) \right). } \]

Define

\[ (V\star_q^QW)_K(h)

:=

\sum_{r=0}^K Q\!\left( V_r(q^{K-r}h), W_{K-r}(h) \right). \]

Then

\[ \boxed{ \mathcal C[Q(F,G)] = \mathcal C[F]\star_q^Q\mathcal C[G]. } \]

This is the genuinely important abstraction:

> BKQR converts a physical bilinear operation into an exact convolution in representation grade.

For ordinary multiplication it is a multiplication-aware Banach algebra. For Navier–Stokes, \(Q\) will eventually become convection or tensor product.

We can also define representation-vanishing ideals. If

\[ \mu(h)\to0, \]

set

\[ \mathcal I_\mu(Y) = \left\{

**F:**

\|F(h)\|_Y\lesssim\mu(h) \right\}. \]

Bounded bilinearity immediately gives

\[ \boxed{ Q(\mathcal I_\mu,\mathcal I_\nu) \subset \mathcal I_{\mu\nu}. } \]

In particular,

\[ Q(\mathcal I_\mu,\mathcal I_\mu) \subset \mathcal I_{\mu^2}. \]

This does **not** need

\[ \mu(h)=h^\alpha. \]

For example,

\[ \mu(h)=\frac1{\log(1/h)} \]

still squares to

\[ \mu(h)^2=\frac1{\log^2(1/h)}. \]

This is why the later Candidate-F argument does not fundamentally require a universal \(h^\alpha\) law. filecite turn0file1

---

# IV. Add an inner product: Hilbert spaces

Now let

\[

**H**

\]

be a Hilbert space with scalar product

\[ \langle u,v\rangle_H. \]

We gain geometry unavailable in Banach spaces:

\[ \|u\|_H^2=\langle u,u\rangle_H, \]

orthogonality,
\[ H=E\oplus E^\perp, \]
self-adjoint operators, spectral projections, positive quadratic forms, and energy spheres.
This is the natural level at which Navier–Stokes energy lives.
Now let
\[ A\ge0 \]
be self-adjoint on \(H\).
Then functional calculus allows
\[ f(A) \]
for suitable scalar functions \(f\).
That one assumption is enough to construct essentially the entire **positive BKQR clock/shell calculus** before mentioning convection.
---
# V. The Hilbert scale generated by \(A\)
From \(A\) define
\[ H_A^s
:=
D(A^{s/2}) \]
in the homogeneous setting, or
\[ \mathcal H_A^s
:=
D((I+A)^{s/2}) \]
inhomogeneously.
Thus
\[ \cdots \subset H_A^2 \subset H_A^1 \subset H \subset H_A^{-1} \subset H_A^{-2} \subset\cdots. \]
This will become the Sobolev hierarchy once \(A\) becomes the Stokes operator.
The critical Navier–Stokes level in three dimensions will be
\[

**H_A^{1/2}.**

\]
---
# VI. Now instantiate the physical 3D Navier–Stokes space
Take
\[ \Omega=\mathbb T^3. \]
Start with smooth mean-zero divergence-free vector fields
\[ \mathcal D_\sigma = \left\{ u\in C^\infty(\mathbb T^3;\mathbb R^3): \nabla\cdot u=0,\, \int_{\mathbb T^3}u\,dx=0 \right\}. \]
Complete in \(L^2\):
\[ \boxed{

**H**

=
L^2_\sigma(\mathbb T^3).
} \]
Complete in \(H^1\):
\[ \boxed{

**V**

=
H^1_\sigma(\mathbb T^3).
} \]
Then we have the Gelfand triple
\[ \boxed{ V\hookrightarrow H \simeq H^* \hookrightarrow V^*. } \]
This is arguably the fundamental functional-analytic skeleton of 3D Navier–Stokes.
The Leray projector is
\[ P:L^2(\mathbb T^3;\mathbb R^3)\to H. \]
In Fourier variables,
\[ P_k = I-\frac{k\otimes k}{|k|^2}, \qquad k\ne0. \]
Therefore each nonzero Fourier mode lives in
\[ \boxed{ k^\perp = \{v\in\mathbb C^3:k\cdot v=0\}. } \]
And in three dimensions,
\[ \boxed{ \dim_\mathbb C k^\perp=2. } \]
This fact becomes enormously important later.
The raw tensor
\[ u\otimes v \]

does indeed live in a $(3\times3)$ tensor space, but the physical nonlinear output is not a generic nine-dimensional tensor:

$$ u\otimes v \;\xrightarrow{\nabla\cdot}\; \mathbb R^3 \;\xrightarrow{P}\; k^\perp. $$

Thus at each Fourier output $(k)$, only a **two-dimensional polarization fibre** survives.

That is the first place where the 3D geometry becomes much smaller than the naive $(3\times3)$ tensor description.

---

# VII. The Stokes operator

Define

$$ \boxed{ A=-P\Delta. } $$

On the torus, for divergence-free fields,

$$ Au=-\Delta u, $$

with

$$ D(A)=H^2_\sigma. $$

It is positive and self-adjoint:

$$ \langle Au,u\rangle = \|\nabla u\|_2^2. $$

Its Fourier eigenvalue is

$$ |k|^2. $$

Consequently,

$$ A^{1/2}\sim |\nabla|, \qquad A^{1/4}\sim |\nabla|^{1/2}. $$

Hence

$$ D(A^{1/4})=H^{1/2}_\sigma $$

up to the usual equivalent torus norms.

---

# VIII. The nonlinear map

Define

$$ \boxed{
B(u,v)=P((u\cdot\nabla)v)
      =P\nabla\cdot(u\otimes v).
} $$

At the energy level,

$$ B:V\times V\to V^*. $$

Define its trilinear form

$$ b(a,v,w) = \langle B(a,v),w\rangle. $$

For divergence-free $(a)$,

$$ b(a,v,w)=-b(a,w,v) $$

and therefore

$$ \boxed{ b(a,v,v)=0. } $$

In particular,

$$ \langle B(u,u),u\rangle=0. $$

This skew cancellation is what produces the physical energy balance.

The target equation is therefore

$$ \boxed{ \partial_tu+\nu Au+B(u,u)=0. } $$

---

# IX. The trajectory space

The ordinary energy-level trajectory belongs to

$$ \boxed{ \mathcal E_T =
L^\infty(0,T;H)
\cap
L^2(0,T;V).
} $$

At the weak level,

$$ u\in C_w([0,T];H) $$

and the equation is interpreted in $(V^*)$.

The energy inequality is

$$ \frac12\|u(t)\|_2^2 + \nu\int_0^t\|A^{1/2}u(s)\|_2^2\,ds \le \frac12\|u_0\|_2^2. $$

For smooth solutions equality holds.

So the standard energy theory controls

$$ L_t^\infty L_x^2 \cap L_t^2 H_x^1. $$

What it does **not** directly control is the critical three-dimensional midpoint

$$
H^{1/2}.
$$

---

# X. The critical space

For the three-dimensional scaling,

$$

$$
H^{1/2}
$$
and
$$
L^3
$$
sit at the critical regularity level.
On $\mathbb T^3$,
$$\boxed{
H^{1/2}\hookrightarrow L^3.
}$$
Therefore a uniform-in-time estimate of the form
$$\sup_{t<T}\|u(t)\|_{H^{1/2}}<\infty$$
or equivalently a suitable $L^\infty_tL^3_x$ bound reaches standard critical regularity theory.
This is why the BKQR shell moment at exponent $1/2$ is not an arbitrary choice.

---

# XI. Put the positive BKQR clock on the Stokes Hilbert space

We now use nothing special about convection—only that $A\ge0$ is self-adjoint.
Define
$$\boxed{ f_q(z) = \prod_{n=0}^\infty(1+q^nz)^{-1}. }$$
It satisfies
$$\boxed{ f_q(qz)=(1+z)f_q(z). }$$
Define
$$\boxed{
J_h=f_q(hA).
}$$
Then functional calculus gives the exact identity
$$\boxed{
J_{qh}=(I+hA)J_h.
}$$
Equivalently,
$$
J_{qh}-J_h=hAJ_h.
$$
There is also the positive probabilistic representation
$$
J_h
= \mathbb E[e^{-hY_\infty A}],
$$
where
$$Y_\infty=\sum_{n\ge0}q^nE_n, \qquad E_n\sim\operatorname{Exp}(1).$$
So the BKQR regularizer is a positive average of Stokes semigroups. filecite turn0file0

---

# XII. BKQR shells on the Stokes spectrum

Apply the correction layers to $J_h$:
$$\boxed{
\Pi_K(hA):=C_KJ_h.
}$$
The scalar multiplier is
$$\boxed{ \pi_K(x) = \frac{q^{K(K-1)/2}x^Kf_q(x)} {(q;q)_K}. }$$
And exactly,
$$\pi_K(x)\ge0,$$
$$\sum_{K=0}^\infty\pi_K(x)=1.$$
Hence
$$\boxed{ \Pi_K(hA)\ge0, \qquad \sum_K\Pi_K(hA)=I. }$$
This is stronger than an arbitrary Littlewood–Paley decomposition: it is a **positive probability partition of spectral space**.
There is also the raising law
$$\boxed{ hA\Pi_K = (q^{-K}-q)\Pi_{K+1}. }$$
So the representation index $K$ becomes a logarithmic physical spectral coordinate. filecite turn0file1

---

# XIII. The shell-number operator

The shell distribution is Bernoulli.
Its mean is
$$g(x) = \sum_{n\ge0} \frac{q^nx}{1+q^nx}.$$
Define
$$\boxed{ N_h=g(hA). }$$

Exactly,
$$ g(x)=-x\partial_x\log f_q(x), $$
and
$$ xg'(x)=\operatorname{Var}(N_x). $$
Moreover,
$$ g(x) = \log_{1/q}x+O_q(1). $$
Therefore $N_h$ acts approximately like
$$ \log_{1/q}(hA) $$
but remains a bounded-difference, positive spectral object.
---
# XIV. BKQR reconstructs the Sobolev hierarchy
Define the shell moment
$$
m_\sigma(x)
= \sum_Kq^{-\sigma K}\pi_K(x) = \frac{f_q(x)}{f_q(q^{-\sigma}x)}. $$
For
$$ 0\le\sigma\le1, $$
the frozen estimate is
$$ \boxed{
m_\sigma(x)\asymp_{q,\sigma}(1+x)^\sigma.
} $$
Consequently
$$
M_\sigma(u)
:=
\langle u,m_\sigma(hA)u\rangle $$
satisfies
$$ \boxed{
M_\sigma(u)
\asymp \|(I+hA)^{\sigma/2}u\|_2^2. } $$
At
$$ \sigma=\frac12, $$
$$ \boxed{
M_{1/2}(u)
\asymp \|u\|_2^2 + h^{1/2}\|A^{1/4}u\|_2^2. } $$
Thus BKQR has reconstructed precisely the critical $H^{1/2}$ level from its positive scale probability law.
filecite turn0file0
---
# XV. Navier–Stokes seen through BKQR shells
For a smooth solution define
$$

**E_K**

$$= \langle u,\Pi_Ku\rangle, $$
$$

**D_K**

$$= \langle A^{1/2}u,\Pi_KA^{1/2}u\rangle, $$
$$

**T_K**

$$= -\langle B(u,u),\Pi_Ku\rangle. $$
Then exactly
$$ \boxed{ \frac12\partial_tE_K+\nu D_K=T_K. } $$
Energy conservation of convection gives
$$ \sum_KT_K=0. $$
The raising relation gives
$$ \boxed{

**D_K**

$$= \frac{q^{-K}-q}{h}E_{K+1}. } $$
For a weighted moment,
$$ T_\sigma = -\langle B(u,u),m_\sigma(hA)u\rangle. $$
Skewness converts this into
$$ \boxed{ T_\sigma = -\frac12 \left\langle [m_\sigma(hA),u\cdot\nabla]u, u \right\rangle. } $$
This is already a major conceptual compression:

> the obstruction to controlling a positive Sobolev scale is exactly the failure of transport to commute with the corresponding BKQR scale measurement.
At \(\sigma=1/2\), this is the critical 3D problem.
---
# XVI. Now add the representation family of PDE solutions
Instead of acting only on one physical solution, define for every \(h>0\)
\[ \boxed{ \partial_tu_h + \nu Au_h +
B(J_hu_h,u_h)=0,
\qquad
u_h(0)=u_0.
} \]
This is the positive regularized family used throughout the source stack.
Since \(J_hu_h\) is divergence-free,
\[ \boxed{ \frac12\frac d{dt}\|u_h\|_2^2 + \nu\|A^{1/2}u_h\|_2^2=0. } \]
So we now have a new mathematical object:
\[ \boxed{ h\longmapsto u_h \in\mathcal E_T. } \]
It is simultaneously
- a curve in a Banach/Hilbert trajectory space, - a solution family of PDEs, - a BKQR representation-scale object.
This is where the two mathematical worlds really meet.
---
# XVII. Apply the geometric annihilator to the PDE family
Take
\[
h_j=hq^j,
\qquad
u_j=u_{h_j},
\]
and define
\[ \boxed{ U_K=\sum_{j=0}^Kc_j^{(K)}u_j. } \]
Set
\[ d_j=u_j-U_K. \]
Since
\[ \sum_jc_j=1, \]
we have
\[ \boxed{ \sum_jc_jd_j=0. } \]
Now apply the quadratic identity.
The extrapolated field satisfies exactly
\[ \boxed{ \partial_tU_K + \nu AU_K +
B(U_K,U_K)
= P\nabla\cdot G_K, } \]
where
\[ \boxed{
G_K
= -\sum_jc_jd_j\otimes d_j + \sum_jc_j[(I-J_j)u_j]\otimes u_j. } \]
Thus
\[
G_K=G_K^{\rm disp}+G_K^{\rm moll}.
\]
And the crucial feature is
\[ \boxed{
G_K^{\rm disp}
= -\sum_jc_jd_j\otimes d_j. } \]
There is **no term linear in \(d_j\)**.
All first-order representation error has disappeared.
That is not a Navier–Stokes estimate. It is an exact consequence of affine BKQR geometry plus quadratic nonlinearity.
filecite turn0file0 filecite turn0file1
---
# XVIII. Candidate F and the scale Green kernel
Write neighboring representation differences
\[ \delta_r=u_{r+1}-u_r. \]
Then
\[
G_K^{\rm disp}
= -\sum_{r,s<K}
M_{rs}^{(K)}
\delta_r\otimes\delta_s \]

with
\[ \boxed{
M_{rs}^{(K)}
= p_{\min(r,s)} \tau_{\max(r,s)}. } \]
The important audit correction was that
\[ \boxed{ \|M^{(K)}\|_{\rm op}\not\to0. } \]
The non-small part instead moves with the finest representation corner.
So BKQR does **not** magically make the quadratic matrix disappear.
But we have the stronger useful statement:
\[ \boxed{ \delta_r\to0 \quad\Longrightarrow\quad
G_K^{\rm disp}\to0
} \]
in every compatible bilinear topology, assuming merely boundedness of the increment sequence.
No rate such as
\[ \delta_r=O(h^\alpha) \]
is necessary.
This is an extremely important point for the architecture. filecite turn0file0
---
# XIX. The covariant derivative in representation space
A raw derivative in \(h\) sees both the physical change of \(u_h\) and the deterministic motion of the BKQR clock.
Set
\[ Z_h=-h\partial_hu_h. \]
Define
\[ K_h=-h\partial_hJ_h. \]
The exact factorization is
\[ K_h=J_hN_h. \]
Let \(L_h\) denote the linearized regularized NSE operator.
Then
\[ L_hZ_h = -B(K_hu_h,u_h). \]
Now define
\[ \boxed{ \Gamma_h = Z_h+N_hu_h. } \]
The deterministic clock component cancels:
\[ \boxed{ L_h\Gamma_h = [C_{J_hu_h},N_h]u_h. } \]
This is essentially the **BKQR covariant derivative** of the PDE trajectory. filecite turn0file0
Its Fourier symbol contains
\[ (r\cdot\widehat a(p)) \left[ g(h|r|^2)-g(h|k|^2) \right], \qquad p+r=k. \]
Immediately,
\[ |r|=|k| \quad\Longrightarrow\quad g(h|r|^2)-g(h|k|^2)=0. \]
So equal-radius transport is invisible to the radial scale connection.
And incompressibility yields
\[ p\cdot\widehat a(p)=0, \]
hence
\[ r\cdot\widehat a(p) = k\cdot\widehat a(p). \]
This is what allows the high-high \(\to\) low derivative to land on the low output rather than remain at the high parent frequency.
---
# XX. What the full abstract structure now looks like
We have built, bottom-to-top,
\[ \boxed{ \text{metric scale orbit} } \]
\[ \downarrow \]
\[ \boxed{ \text{Banach geometric annihilation} } \]
\[ \downarrow \]
\[ \boxed{ \text{bilinear }q\text{-convolution algebra} } \]
\[ \downarrow \]
\[ \boxed{ \text{Hilbert spectral functional calculus} } \]
\[ \downarrow \]
\[ \boxed{ \text{positive BKQR clock and shells} } \]
\[ \downarrow \]
\[ \boxed{
L^2_\sigma
\supset
H^1_\sigma,\quad
A=-P\Delta } \]
\[ \downarrow \]
\[ \boxed{
B(u,v)=P\nabla\cdot(u\otimes v)

} \]
\[ \downarrow \]
\[ \boxed{ h\mapsto u_h } \]
\[ \downarrow \]
\[ \boxed{ \text{Candidate-F quadratic residual} } \]
together with the parallel route
\[ \boxed{ \text{shell moments} \to
H^{1/2}
\to \text{critical commutator}. } \]
That is, in my view, the correct common trunk before we branch.
---
# XXI. Five branches I would preserve separately
I would **not merge these yet**.
## Path A — Positive critical shell flux
Stay as close to the physical NSE as possible.
The central object is
\[

### T_{1/2}

= -\frac12 \langle [m_{1/2}(hA),u\cdot\nabla]u,u \rangle. \]
The eventual target is a strict viscous estimate such as
\[ \boxed{

### |T_{1/2}|

\le \theta\nu D_{1/2} +

### C(1+M_1)M_{1/2},

\qquad \theta<1. } \]
The later source work already supplies potentially useful local ingredients:
\[ \text{equal-radius nullity}, \]
\[ \text{homochiral critical nullity}, \]
\[ \text{heterochiral radial susceptibility}, \]
\[ \text{Poisson micro-shell radial gain}, \]
and sparse/C4-type gains.
The real unresolved part is the **radially non-null, soft-shell, large-data critical transfer**. filecite turn0file4
This is the most direct path to regularity.
---
## Path B — Candidate F → small residual → certificate
Here the objective is not initially to bound the physical critical moment.
Instead prove
\[ \boxed{
u_{qh}-u_h\to0
} \]
in some trajectory space \(Y_T\).
Then the no-rate Candidate-F theorem gives
\[
G_K^{\rm disp}\to0
\]
in a compatible stress space \(X_T\).
If simultaneously
\[
G_K^{\rm moll}\to0,
\]
then
\[ \boxed{
G_K\to0.
} \]
So \(U_K\) becomes an increasingly accurate solution of the actual unregularized NSE.
One can then seek either an a-posteriori regularity certificate directly or a nonlinear Newton-type residual squaring scheme.
The hard question here is extremely concrete:
> Find one pair \(Y_T,X_T\) in which neighboring-scale continuity, the quadratic tensor map, the mollifier defect, the Stokes response, and the final regularity certificate all coexist without circularity.
This is the cleanest **residual/certificate route**. filecite turn0file1
---
## Path C — Covariant/front response
Rather than assuming neighboring-scale continuity, try to derive it dynamically.

Use
$$ L_h\Gamma_h = [C_{J_hu_h},N_h]u_h $$
or the later bounded front connection
$$ P_h = hA(I+hA)^{-1}. $$
The front formulation has a forcing symbol of schematic form
$$ \boxed{ |\sigma_h(p,r,k)| \lesssim \min(|p|,|r|,|k|) \frac{hH^2}{(1+hH^2)^2}. } $$
This simultaneously produces
$$ \text{minimum-frequency derivative gain} $$
and
$$ \text{localization near }hH^2\sim1. $$
So far interactions and interactions far from the moving face should be perturbative.
The obstruction is compressed into
$$ \boxed{ |p|\sim|r|\sim|k|, \qquad h|p|^2\sim1. } $$
Then either response decays and gives Q2, or failure produces an adjoint resonance to classify. This is probably the cleanest **analytic engine** for Path B, but I would still keep it conceptually separate for now.
filecite turn0file2

---

## Path D — 3D quotient/polarization geometry

This path uses something specifically three-dimensional.
At each output,
$$ k^\perp $$
has dimension $2$.
Define the span of active nonlinear response vectors
$$ S_u(k)\subset k^\perp. $$
In 2D, $k^\perp$ is one-dimensional, so any nonzero active response already spans the whole output fibre.
In 3D, one can have
$$ \boxed{ \dim S_u(k)=1<2. } $$
Thus the active representation cokernel
$$ Q_{\rm rep}(k) = k^\perp/S_u(k) $$
can itself be one-dimensional.
This is the genuinely new 3D local defect.
The later work attacks this one-dimensional obstruction through several compatible geometries:
$$ \text{plane synchronization}, $$
$$ \text{shear-null sectors}, $$
$$ \text{Beltrami/single-helicity nullity}, $$
$$ \text{Cauchy--Binet costed accessibility}, $$
$$ \text{C4 incidence}, $$
$$ \text{projective holonomy}. $$
The desired theorem is roughly
$$ \boxed{ \text{rank opening} \quad\text{OR}\quad \text{null/synchronized geometry} \quad\text{OR}\quad \text{sparsity}. } $$
This is an attractive path because it tries to explain **why the 3D obstruction is only one extra degree of freedom relative to 2D**, rather than treating the entire $3\times3$ nonlinear tensor as irreducible. filecite turn0file5 filecite turn0file6 filecite turn0file7

---

## Path E — moving-face profile / resonance classification

Suppose Path C fails to establish scale continuity directly.
The source of bad response is already forced toward
$$ hA\sim1. $$
Rescale around that face:
$$ y=\frac{x-x_0}{\sqrt h}, \qquad \tau=\frac{t-t_0}{h}, $$
$$ \boxed{ V_h(y,\tau) = \sqrt h\,
u_h(x_0+\sqrt h\,y,t_0+h\tau).
} $$
Then the representation parameter disappears from the leading PDE:
$$ \partial_\tau V + \nu A_yV + B_y(J_1V,V)=0. $$
So a Q2 failure is converted into a **fixed-scale profile problem**.
The profile branch then asks:
$$ \text{Does the profile remain bounded?} $$
$$ \text{Does exterior forcing vanish?} $$
$$ \text{Does it become lower-dimensional?} $$
$$ \text{Does an ancient profile survive?} $$
$$ \text{Does second representation curvature persist?} $$
The more recent anisotropic work introduces a genuinely local face intensity $\Theta_\Lambda$, obtains local anisotropic mass from dangerous transfer, kills bounded autonomous ancient face profiles using the spectral gap, and routes high-intensity profiles toward inviscid lower-dimensional dynamics or retained exterior forcing.
filecite turn0file3

This is the natural **compactness/obstruction branch** if direct estimates fail.

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# XXII. The trunk I would freeze before branching

So for our main conversation I would currently freeze only this common architecture:

$$\boxed{ \begin{array}{c} \text{Metric scale geometry}\\ \downarrow\\ \text{Banach BKQR annihilator}\\ \downarrow\\ \text{exact bilinear }q\text{-convolution}\\ \downarrow\\ \text{Hilbert/Stokes functional calculus}\\ \downarrow\\ \text{positive clock }J_h +\text{ positive shells }\Pi_K\\ \downarrow\\ \text{regularized NSE family }u_h\\ \downarrow\\ \text{exact Candidate-F quadratic residual} \end{array} }$$

and beside it,

$$\boxed{ \begin{array}{c} \Pi_K\\ \downarrow\\ m_{1/2}(hA)\\ \downarrow\\ H^{1/2}\leftrightarrow L^3\\ \downarrow\\ \text{critical commutator} \end{array} }$$

and finally the representation differential geometry

$$\boxed{ \begin{array}{c} h\mapsto u_h\\ \downarrow\\ Z_h=-h\partial_hu_h\\ \downarrow\\ \Gamma_h=Z_h+N_hu_h\\ \downarrow\\ L_h\Gamma_h = [C_{J_hu_h},N_h]u_h . \end{array} }$$

**That is the common trunk.**

From there I would let the five branches live independently:

$$\boxed{ \text{A: critical shell flux} }$$

$$\boxed{ \text{B: Candidate-F certificate} }$$

$$\boxed{ \text{C: covariant/front response} }$$

$$\boxed{ \text{D: quotient/null-rich-sparse geometry} }$$

$$\boxed{ \text{E: moving-face profiles/resonances}. }$$

There are already obvious bridges between them—but I agree with your instinct **not to merge those bridges yet**. Each path should first be allowed to develop far enough that we can see which theorem is genuinely carrying new information and which path is merely re-expressing another one.

The next branch I would personally explore first from this trunk is **Path D from the purely dimensional viewpoint**: start with $d=2$, prove exactly why the active quotient disappears there, then increase $d$ to $3$ and characterize precisely the single extra quotient line. That may tell us what object Paths A–C actually have to control, before we try to estimate it.